



Faculty of Computer Studies
Information Technology and Computing Department

Automated Virtual SOC Environment

TM471-B: Part Two of Graduation Project

2025/2026

Submitted to

Arab Open University

Supervised By

Dr-Ibrahim Ahmed Hassan

Presented by

Mostafa Badawy Mohamed Badawy

Declaration of No Plagiarism

I hereby declare that this submitted report work is a result of my own efforts and I have not plagiarized any other person's work. I have provided all references for information that I have used and quoted in my work.

Name of Student: **Mostafa Badawy Mohamed**

Mostafa Badawy

تخليد ذكرى الدكتور الفاضل محمود عطالله

بسم الله الرحمن الرحيم

﴿ يَرْفَعُ اللَّهُ الَّذِينَ آمَنُوا مِنْكُمْ وَالَّذِينَ أُوتُوا الْعِلْمَ دَرَجَاتٍ ﴾

﴿ كُلُّ نَفْسٍ ذَائِقَةُ الْمَوْتِ ﴾

بقلوب يملؤها الحزن، ورضاً بقضاء الله وقدره، نُهدي هذا العمل العلمي إلى روح الأستاذ الجليل الدكتور محمود عطالله ، الذي لم يكن مجرد أستاذ جامعي، بل قدوة في الأخلاق، ورمزاً للعطاء، وحاملاً للعلم رسالة وإنسانية. كان رحمه الله مثلاً للأستاذ الحقيقي؛ بابتسامته الصادقة، وصدرة الرحب لطلابه، وحرصه الدائم على ألا يترك طالباً دون توجيه أو دعم. بذل وقته وجهده بإخلاص، وكان يفرح بنجاح طلابه كأنه نجاحه الشخصي. وإذ نقدّم هذا المشروع، نقف وقفة وفاء وامتنان لمن علّمنا وأرشدنا، وغرس فينا حبّ العلم والسعي الدائم للمعرفة. لقد رحل الجسد، لكن الأثر باقٍ، والعلم شاهد، والذكرى حيّة في القلوب. نسأل الله أن يجعل هذا العمل في ميزان حسناته، وأن يغفر له ويرحمه، ويجعل علمه صدقة جارية، ويجمعه مع الصالحين، ويلهم أهله ومحبيه الصبر والسلوان. رحمك الله يا دكتور محمود، وسيبقى ذكرك طيباً ما بقي العلم وأهله.



Abstract

This work explores the creation of a simulated Security Operations Center environment using freely available software components. Rather than relying on proprietary security platforms, the system brings together multiple open-source solutions to support event visibility, alert handling, and response coordination. The approach is particularly suitable for learning environments and small teams where financial and infrastructure limitations prevent the use of enterprise security products.

Log data is collected and examined through a centralized analytics framework, while incident tracking and investigative collaboration are handled using a dedicated case management tool. Automated actions are coordinated through a workflow engine that links detection events with predefined responses. To produce realistic security activity, a Windows-based system generates endpoint telemetry and controlled attack scenarios are executed to test detection logic. Notifications are delivered instantly to analysts through a messaging service when events occur. The implementation demonstrates that practical SOC capabilities can be achieved without commercial tools, supporting both skill development and limited operational needs.

Acknowledgements

I would like to express my sincere appreciation to my supervisor, **Dr-Ibrahim Ahmed Hassan**, for his ongoing guidance and constructive feedback throughout this project. His expertise greatly contributed to shaping the technical direction and academic structure of this work.

I am equally thankful to my colleague, Abdelrahman Saad, for his cooperation, thoughtful discussions, and support during the configuration and testing phases. His contributions helped enhance the overall quality and outcomes of the project.

Finally, I extend my appreciation to the Arab Open University for offering the resources, facilities, and academic environment that enabled me to successfully complete this work.

Table of contents

Implementation of an Automated Virtual SOC Environment

Abstract	iii
Acknowledgements	iv
Table of contents	v
List of figures and Diagrams:	vii

Chapter One: Introduction 1

1.1 Overview	2
1.2 Motivations of the project	3
1.3 Problems of the project	3
1.4 Aims and Objectives	4
1.5 Scope and Constraints of the project	4
1.6 Suggested solution	5
1.7 Project Plan	8

Chapter Two: Literature Review 9

2.1 Background Information	10
2.2 Importance of Virtual SOC Platforms in Education and Industry	11
2.3 Similar applications	12
2.3.1 Splunk Enterprise Security	11
2.3.2 Palo Alto Cortex XSO	14
2.3.3 MITRE CALDERA	15
2.4 Disadvantages of similar applications	16
2.5 Related Works – Summary and Comparison	17
2.6 Comparison with My Project	18

Chapter Three: Requirements and Analysis 19

3.1 Functional Requirements	20
3.2 Non-Functional Requirements	21
3.3 System Workflow	22
3.4 Software Tools & Hardware Specifications	23
3.4.1 Software Requirements:	24
3.4.2 Hardware Requirements:	25
3.5 Project-Specific Code of Ethics – Virtual SOC Project:	26
3.6 Analysis Diagrams	27
3.6.1 Use Case Diagram	28
3.6.2 Sequence diagram	29
3.6.3 State Diagram	30
3.6.4 Block Diagram	31
3.6.5 Activity Diagram	32

Chapter Four: Design, Implementation and Testing	33
4.1 System Architecture.....	34
4.2 Infrastructure Setup.....	35
4.3 Workflow Implementation.....	36
4.3.1 Workflow 1: Threat Detection and Enrichment	37
4.3.2 Workflow 2: Phishing Email Detection	38
4.3.3 Workflow 3: Endpoint Isolation and Malware Containment	39
4.4 System Integration	40
4.5 Challenges and Solutions.....	41
 Chapter Five: Results and Discussion.....	 42
5.1 Overview of Findings	43
5.2 Goals Achieved	44
5.3 Testing Scenarios	45
5.3.1 Workflow 1: IP Threat Intelligence and Incident Management	46
5.3.2 Workflow 2: Phishing Email Detection	47
5.3.3 Workflow 3: Endpoint Isolation and Malware Containment	48
5.4 Performance Evaluation.....	49
5.5 Limitations	50
5.6 Future Work	51
5.7 Ethical and Legal Considerations	52
 Chapter Six: Conclusion	 53
6.1 Summary	54
6.2 Key Achievements	55
6.3 Broader Significance.....	56
6.4 Limitations and Future Directions	57
6.5 Final Remarks	58
References	59

List of figures and Diagrams:

Figure 1- gannt chart for project plan	1
Figure 2- Splunk Website	2
Figure 3- PaloAltonetworks website.....	3
Figure 4- caldera.mitre website.....	4
Figure 5 – Workflow 1: Threat Detection and Enrichment.....	5
Figure 6 – Workflow 2: Phishing Email Detection.....	6
Figure 7 – Workflow 2: Endpoint Isolation and Malware Containment.....	7
Figure 8 – Telegram SOC Alert (WF1).....	8
Figure 9 – The Hive Case (WF2).....	9
Figure 10 – Email and Telegram Alerts (WF2).....	10
Figure 11 – Malware Alert in Kibana (WF2).....	11
Figure 12 – FortiGate Policy (WF3).....	12
Figure 13 – Telegram MALWARE DETECTED Alert (WF3).....	13
Figure 19 – Email MALWARE DETECTED Alert (WF3).....	14
Figure 14 – SSH Scan Detected Alert (WF3).....	15
Figure 15 – FortiGate Policies (WF3).....	16
Figure 16 – Telegram NETWORK ATTACK DETECTED Alert (WF3).....	17
Figure 17 – Email NETWORK ATTACK DETECTED Alert (WF3).....	18
Figure 18 – Telegram IP Already Blocked Alert (WF1).....	19
Figure 19 – Telegram REPEAT ATTACKER DETECTED Alert (WF3).....	20
3.6.1 Use Case Diagram.....	28
3.6.2 Sequence diagram	29
3.6.3 State Diagram.....	30
3.6.4 Block Diagram	31
3.6.5 Activity Diagram	32

Chapter One: Introduction

1.1 overview

Security Operations Centers (SOCs) are essential components of modern cybersecurity infrastructure. They provide organizations with the capability to continuously monitor networks, detect malicious activity, and coordinate responses to security incidents. Despite their importance, building and maintaining a traditional SOC requires significant financial investment, specialized hardware, and a team of experienced personnel resources that are often unavailable to academic institutions, small organizations, and individual learners.

This project addresses that gap by designing and implementing a virtual SOC environment built entirely on open-source technologies. The system replicates the core functions of a real-world SOC, including network traffic monitoring, intrusion detection, threat intelligence enrichment, automated incident response, and analyst notification all within a controlled and reproducible lab setup.

The core of the system is built around the Elastic Stack, which serves as the SIEM platform responsible for log ingestion, storage, search, and visualization. Security Onion is used as the deployment environment for the Elastic Stack, providing an integrated setup that bundles Zeek for network traffic analysis and Suricata for intrusion detection alongside the ELK components reducing configuration overhead while delivering a fully functional ELK environment in a single deployment. The automation layer is implemented through n8n, which orchestrates three dedicated workflows: Workflow 1 for Incident Management and IP Threat Intelligence, Workflow 2 for Phishing Email Detection, and Workflow 3 for Endpoint Isolation and Malware Containment. TheHive provides structured case management, FortiGate handles automated network enforcement, and analysts receive real-time notifications through Telegram and Gmail.

Unlike theoretical SOC models, this project provides a fully functional and practical implementation operating against live simulated attack traffic, enabling dynamic testing and genuine end-to-end validation of detection and response capabilities.

1.2 Motivations of the project

The motivation for this project stems from two parallel challenges facing cybersecurity education and small organizations alike. First, commercial SIEM and SOAR platforms — such as Splunk Enterprise Security and Palo Alto Cortex XSOAR — are prohibitively expensive for academic use and require infrastructure that most institutions cannot provide. Second, there is a persistent gap between theoretical cybersecurity knowledge and the practical skills required to operate a real SOC. Students who complete coursework in network security often have little or no experience interpreting live alerts, triaging incidents, or executing automated response workflows.

Open-source tooling has matured significantly in recent years, making it possible to assemble a capable SOC pipeline without commercial licenses. However, most existing implementations focus on isolated components rather than delivering a fully integrated environment. This project aims to close that gap by providing a cohesive, end-to-end platform that students and small security teams can deploy, test, and learn from.

1.3 Problems of the project

Traditional SOC deployments present three core challenges that limit their use in educational and resource-constrained settings.

The first is cost: enterprise SIEM and SOAR solutions require licensing fees that are out of reach for most academic programs and small businesses.

The second is complexity: integrating detection tools, automation engines, and case management platforms typically requires deep expertise and significant configuration effort.

The third is realism: most training environments use static log files rather than live attack simulations, which prevents learners from experiencing the dynamic nature of real incident response.

Together, these challenges mean that students and small teams are frequently left with either no hands-on SOC experience or with fragmented setups that do not reflect how a production SOC actually operates. There is a clear need for an integrated, cost-free, and realistic SOC environment that can be deployed on modest hardware and used for both training and limited operational purposes.

1.4 Aims and Objectives

The primary aim of this project is to design and implement a fully functional virtual SOC environment using open-source tools, capable of real-time detection, automated enrichment, and coordinated incident response.

The specific objectives are:

- To deploy the Elastic Stack as the core SIEM platform for log ingestion, analysis, and alert generation, using Security Onion as the integrated deployment environment.
 - To implement three automated n8n workflows: Workflow 1 for Incident Management and IP Threat Intelligence, Workflow 2 for Phishing Email Detection, and Workflow 3 for Endpoint Isolation and Malware Containment.
 - To integrate TheHive for structured case creation and incident tracking.
 - To incorporate FortiGate as an automated enforcement layer for network blocking via REST API.
 - To enrich security events using external threat intelligence services including AbuseIPDB, VirusTotal, GreyNoise, and URLScan.io.
 - To simulate realistic attack scenarios and validate detection and response capabilities end-to-end.
 - To deliver real-time analyst notifications through Telegram and Gmail.
-

1.5 Scope and Constraints of the project

The project is scoped to a single-analyst, virtualized lab environment. It demonstrates the full SOC lifecycle — from traffic capture and alert generation through enrichment, case creation, automated blocking, and notification — within a controlled and isolated network segment.

The following constraints apply:

- The environment runs on a local workstation and is not designed for production-scale deployments or multi-tenant use.
- System performance is bounded by available hardware.
- Some enrichment steps depend on external APIs, meaning that internet connectivity is required for VirusTotal, AbuseIPDB, GreyNoise, and URLScan.io lookups.
- Detection is rule-based and does not include machine learning or behavioral anomaly detection in this iteration.

1.6 Suggested solution

The proposed solution is a fully automated SOC pipeline in which the Elastic Stack continuously stores and indexes network events generated by Zeek and Suricata running within Security Onion. ElastAlert evaluates stored events against predefined detection rules and forwards triggered alerts to n8n via HTTP webhooks. Three dedicated n8n workflows then process each alert category independently.

Workflow 1 — Incident Management and IP Threat Intelligence

Handles external IP-based threats. It enriches source IP addresses through AbuseIPDB, VirusTotal, and GreyNoise, calculates a composite risk score, and for high-risk indicators applies iptables blocking on the Security Onion host, creates a TheHive case, and dispatches full SOC alerts via Telegram and Gmail.

Workflow 2 — Phishing Email Detection

Runs on a scheduled trigger and processes email records stored in Elasticsearch. It applies heuristic detection logic, enriches flagged URLs through VirusTotal and URLScan.io, creates TheHive cases for confirmed phishing emails, and notifies the analyst.

Workflow 3 — Endpoint Isolation and Malware Containment

Handles internal endpoint threats. When an internal host is detected performing brute-force attacks or resolving malware-associated domains, n8n calls the FortiGate REST API to isolate the endpoint at the firewall level and notifies the analyst immediately.

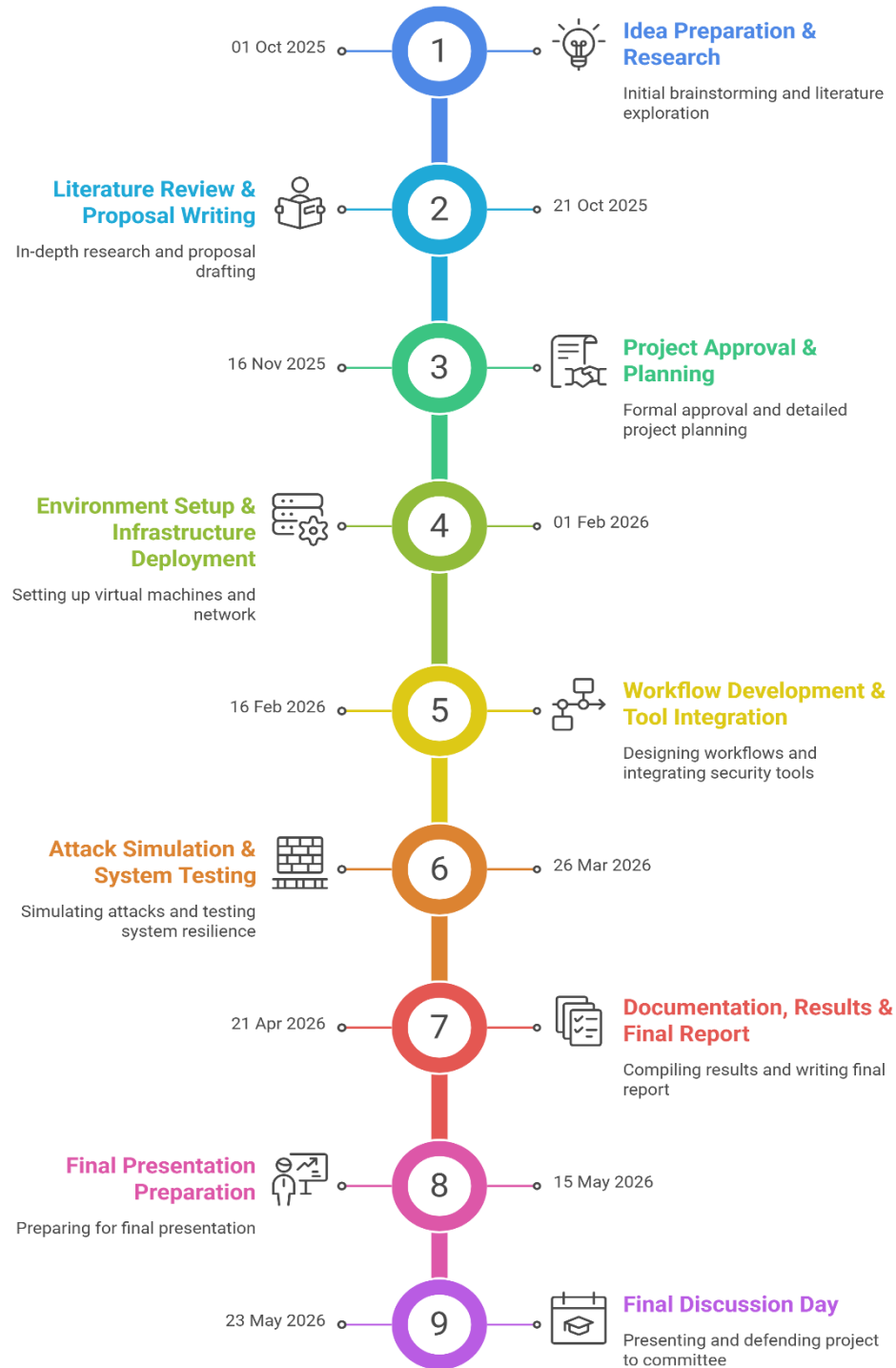
All confirmed high-risk events result in a TheHive case and analyst notification, giving the SOC team a complete picture of each incident alongside clear next steps for investigation.

1.7 Project Plan

Phase	Task	Start Date	End Date	Duration
Part 1	Idea Preparation & Research	01 Oct 2025	20 Oct 2025	3 Weeks
Part 1	Literature Review & Proposal Writing	21 Oct 2025	15 Nov 2025	4 Weeks
Part 1	Project Approval & Planning	16 Nov 2025	30 Nov 2025	2 Weeks
Part 2	Environment Setup & Infrastructure Deployment	01 Feb 2026	15 Feb 2026	2 Weeks
Part 2	Workflow Development & Tool Integration	16 Feb 2026	25 Mar 2026	6 Weeks
Part 2	Attack Simulation & System Testing	26 Mar 2026	20 Apr 2026	4 Weeks
Part 2	Documentation, Results & Final Report	21 Apr 2026	14 May 2026	3 Weeks
Final Stage	Final Presentation Preparation	15 May 2026	22 May 2026	1 Week
Final Milestone	Final Discussion Day	23 May 2026	23 May 2026	Milestone

1.7.1 Project Plan

Implementation of an Automated Virtual SOC Environment



Chapter Two: Literature Review

2.1 Background Information

Security Operations Centers have been a recognized best practice in enterprise cybersecurity for over two decades. Foundational guidance from NIST, including Special Publication 800-61, emphasizes the importance of centralized log collection, event correlation, and structured incident handling procedures. In parallel, the growth of open-source tooling has made it increasingly feasible to replicate SOC capabilities outside the enterprise context.

The Elastic Stack — comprising Elasticsearch for storage and search, Kibana for visualization, and associated ingest and alerting tools — has become a widely adopted platform for real-time log analytics and security event management. Security Onion packages the Elastic Stack together with Zeek and Suricata into a single deployable platform, significantly lowering the barrier to entry for organizations building detection capabilities on top of ELK. TheHive provides a purpose-built case management interface that allows analysts to collaborate, track observables, and follow consistent investigation workflows. The n8n automation engine enables the construction of complex, event-driven workflows that integrate diverse security tools through standard APIs and webhooks.

Attack simulation frameworks such as MITRE's Atomic Red Team implement adversary behaviors drawn from the ATT&CK framework, providing a structured and repeatable method for generating realistic security telemetry and validating detection rules against live traffic.

2.2 Importance of Virtual SOC Platforms in Education and Industry

For Learners

Hands-on simulation environments are substantially more effective than passive instruction. Working analysts must be able to interpret live alert streams, distinguish true positives from false positives, escalate incidents appropriately, and understand how automated workflows interact with manual investigation. These competencies cannot be developed through coursework alone.

Virtual environments that produce real telemetry from simulated attacks allow learners to practice the full analyst workflow — from initial triage through enrichment, case creation, and response — in a safe and repeatable setting.

For Small Organizations

The ability to deploy a capable detection and response pipeline using open-source tools represents a meaningful improvement in security posture without the budget commitments of commercial platforms.

The system developed in this project addresses both audiences: it is accessible enough for academic use and functionally complete enough to support limited operational deployment.

2.3 Similar applications

2.3.1 Splunk Enterprise Security

Splunk Enterprise Security is a leading commercial SIEM platform used by large organizations to collect, index, and search security event data at scale. It offers extensive correlation rules, a large library of prebuilt dashboards, and deep integration with third-party security products.

Its limitations in the context of this project are primarily economic: licensing costs are substantial, and the platform requires dedicated infrastructure that is not feasible for academic or small-team deployments. Splunk also lacks native attack simulation capabilities and does not include open-source automation tooling comparable to n8n.

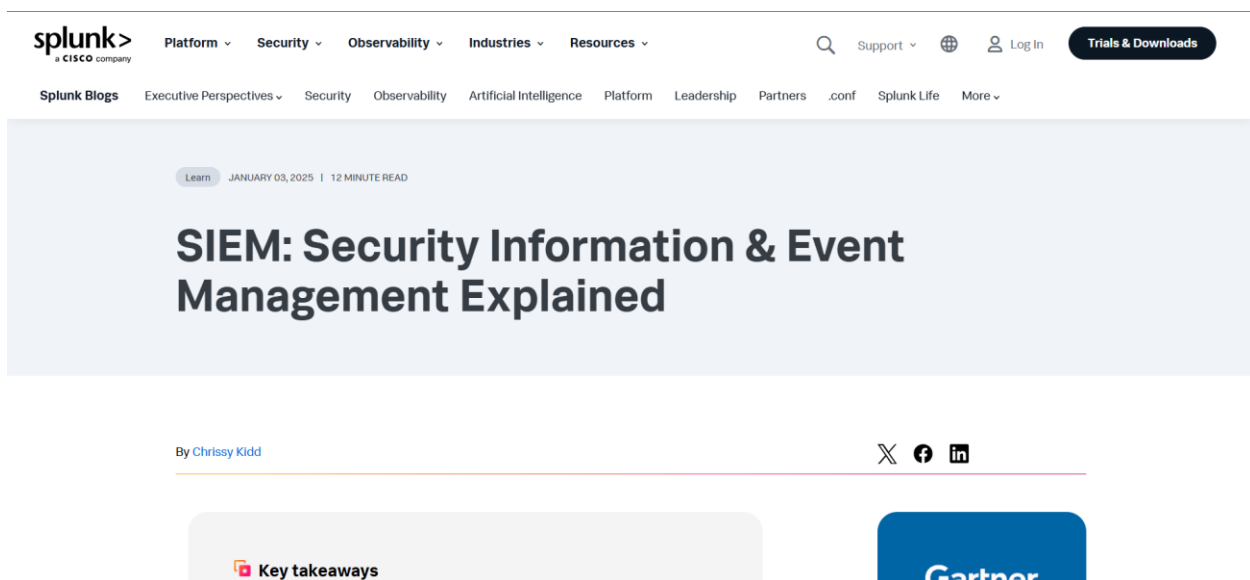


Figure 2 – Splunk Website

Reference: https://www.splunk.com/en_us/solutions/siem.html

2.3.2 Palo Alto Cortex XSOAR

Cortex XSOAR is a commercial SOAR platform designed for enterprise-scale incident response orchestration. It supports sophisticated playbook development, broad integration with security products, and multi-analyst collaboration.

Like Splunk, its commercial licensing and infrastructure requirements place it out of reach for academic environments. Additionally, Cortex XSOAR does not natively integrate with open-source SIEM tools or include attack simulation components, making it less suitable

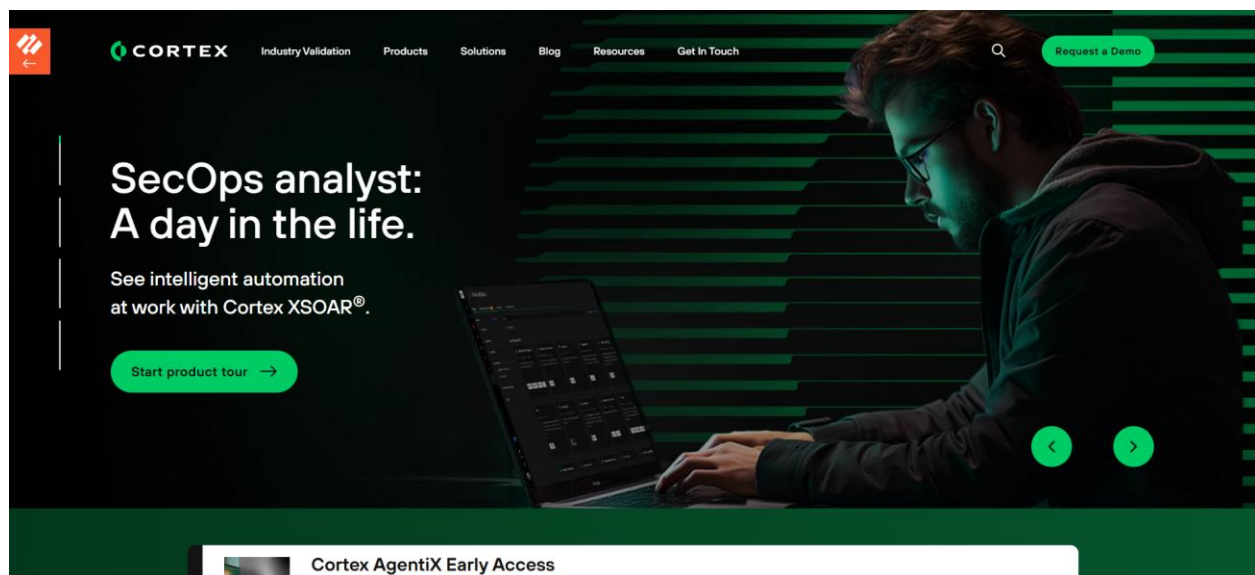


Figure 3- PaloAltonetworks website

Reference: <https://www.paloaltonetworks.com/cortex/cortex-xsoar>

2.3.3 MITRE CALDERA

MITRE CALDERA is an open-source adversary emulation framework that automates attack sequences based on ATT&CK techniques. It is valuable for testing detection rules and assessing defensive coverage, but it does not provide SIEM functionality, case management, or automated response capabilities.

This project uses Atomic Red Team for attack simulation, which complements the integrated detection and response architecture rather than replacing it.

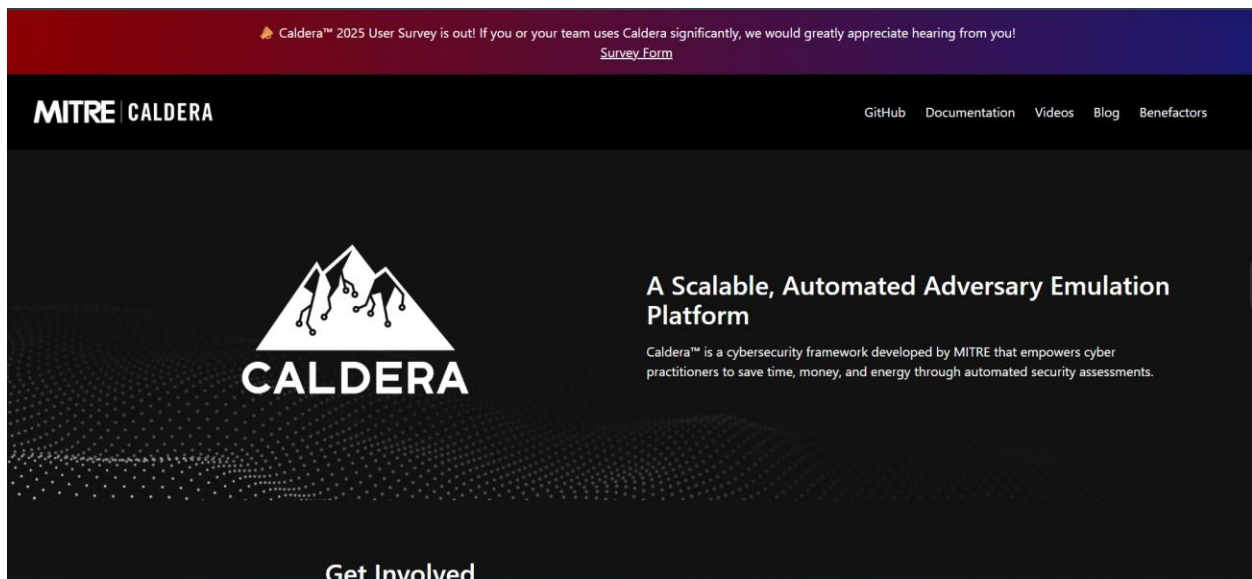


Figure 4 – caldera. miter website

References:

<https://attack.mitre.org/>
<https://caldera.mitre.org/>

2.4 Disadvantages of similar applications

Existing platforms — whether commercial or open-source — share several limitations that motivate the current project.

- Commercial platforms are inaccessible due to cost.
 - Open-source tools tend to address individual components but not all three in an integrated pipeline.
 - Static training datasets remain common in academic implementations, limiting the realism of the learning experience.
 - Automated enforcement actions such as firewall blocking are rarely included in open-source lab implementations.
-

2.5 Related Works – Summary and Comparison

Despite the availability of various cybersecurity platforms, no existing open-source implementation delivers a fully integrated pipeline combining ELK-based detection, multi-source threat intelligence enrichment, automated case management, and firewall-level enforcement in a single deployable system.

Most academic SOC implementations stop at alert generation — they do not close the loop with automated blocking, structured case creation, and multi-channel analyst notification. This project fills that gap.

Reference: https://www.splunk.com/en_us/solutions/siem.html

– Palo Alto Cortex XSOAR (SOC Automation)

- SOAR platform that automates alert triage, ticketing, and incident response coordination.
- Designed for large enterprises with complex infrastructures.
- Not intended for academic experiments or small lab environments.

Reference:

<https://www.paloaltonetworks.com/cortex/cortex-xsoar>

– MITRE CALDERA (Adversary Simulation)

- Tool for simulating realistic attacks based on the MITRE ATT&CK framework to test detection capabilities.
- Results are often evaluated manually after the attack is carried out.
- It does not include direct automated response or complete SIEM integration.

References:

<https://attack.mitre.org/>

<https://caldera.mitre.org/>

2.6 Comparison with My Project

2.6.1 Integrated End-to-End Workflow

Previous implementations addressed SIEM, automation, or attack simulation individually. This project combines all three into one platform:

- Log collection, storage, and analysis using the Elastic Stack as the core SIEM deployed through Security Onion, which bundles Zeek, Suricata, and ELK components into a single environment.
 - Attack simulation using Atomic Red Team and Hydra against live targets.
 - Automated alert handling through three dedicated n8n workflows.
-

2.6.2 Live Detection Testing

Previous studies mostly relied on historical log data or manual assessment. This project enables dynamic, real-time testing of detection rules against live simulated attacks, producing genuine telemetry rather than replayed events.

2.6.3 Cost-Effective and Educational

Previous solutions are either expensive and commercial such as Splunk and Cortex XSOAR or have limited integration scope, like CALDERA.

This project is fully open source, runs on a single workstation, and is designed to be reproducible in academic lab settings.

2.6.4 Customizable Automation

Most open-source SOC implementations stop at alert generation.

This project extends the pipeline to include actual enforcement actions:

- iptables blocking for external IP threats in Workflow 1.
 - FortiGate REST API-based network isolation for internal endpoint threats in Workflow 3.
-

Chapter Three: Requirements and Analysis

3.1 Functional Requirements

The system is required to perform the following core functions:

- Event Ingestion

The Elastic Stack, deployed through Security Onion, continuously stores network events generated by Zeek and Suricata. ElastAlert monitors stored events against detection rules and forwards triggered alerts to n8n via HTTP webhook.

- Data Extraction and Validation

Each incoming event is parsed to extract relevant fields — source IP address, timestamp, alert type, and associated metadata. Events missing required fields are discarded before processing continues.

- Indicator Enrichment

Extracted indicators of compromise — IP addresses, URLs, and domains — are submitted to AbuseIPDB, VirusTotal, GreyNoise, and URLScan.io. Results are consolidated into a single enrichment record per indicator.

- Risk Scoring

A composite risk score is calculated from enrichment data using a weighted model that accounts for VirusTotal malicious engine detections, AbuseIPDB confidence percentage, GreyNoise classification, and Tor exit node status. The score determines downstream routing.

- Automated Response

High-risk indicators trigger automated actions appropriate to the threat type.

- Workflow 1 applies iptables blocking for confirmed external IP threats.
- Workflow 2 creates phishing cases in TheHive.
- Workflow 3 calls the FortiGate REST API to isolate internal endpoints or block malicious domains.

- Duplicate Suppression

Before executing any blocking action, each workflow queries Elasticsearch to determine whether the indicator has been actioned in a previous cycle. Duplicate entries are suppressed and replaced with a low-priority analyst notification.

- Case Creation

All high-risk events confirmed by Workflow 1 and Workflow 2 generate a structured TheHive case containing the full enrichment report, severity classification, assigned tags, and an investigation task with analyst next steps.

- Analyst Notification

Telegram and Gmail alerts are dispatched for every actionable event across all three workflows, including:

- High-risk confirmations.
 - False positive dismissals.
 - Repeat attacker detections.
-

3.2 Non-Functional Requirements

- Performance

Each workflow must complete processing and deliver analyst notifications within a timeframe consistent with near real-time operation.

- Reliability

Workflows must handle malformed or incomplete event data without crashing. Error conditions should produce a notification rather than result in silent failure.

- Scalability

The architecture should support the addition of new workflow types and data sources without requiring modification to existing workflows.

- Security

All inter-component communication should use authenticated API calls. Sensitive values such as API keys and webhook tokens should be stored as environment variables.

- Maintainability

Detection rules, risk scoring logic, and notification templates should be modifiable without requiring structural changes to the workflows themselves.

3.3 System Workflow

The SOC pipeline is implemented as three distinct n8n workflows. All three share a common structural pattern of event ingestion, data normalization, indicator extraction, and coordinated response. Each workflow targets a specific threat category and operates independently.

Workflow 1 — Incident Management and IP Threat Intelligence

Triggered by ElastAlert rules detecting suspicious external network activity. It enriches source IP addresses through multiple threat intelligence APIs, calculates a composite risk score, and for confirmed high-risk indicators:

- Applies iptables blocking.
- Records the blocked IP in Elasticsearch.
- Creates a TheHive case.
- Dispatches full SOC alerts.

Workflow 2 — Phishing Email Detection

Runs on a scheduled trigger and processes email records stored in an Elasticsearch index. It:

- Applies heuristic detection logic to identify phishing indicators.
- Enriches flagged URLs through VirusTotal and URLScan.io.
- Creates TheHive cases for confirmed threats.
- Notifies the analyst.

Workflow 3 — Endpoint Isolation and Malware Containment

Triggered by ElastAlert rules detecting internal endpoint threats — specifically:

- An internal host performing SSH brute-force attacks against other network hosts.
- An internal host resolving domains associated with malware delivery.

Unlike Workflow 1, it does not perform multi-source enrichment; the ElastAlert rule match is itself sufficient justification for immediate isolation. Enforcement is applied via the FortiGate REST API.

3.4 Software Tools & Hardware Specifications

3.4.1 Software Specifications

Component	Specification
Host Operating System	Windows 10 (endpoint machine)
SIEM Platform	Elastic Stack — Elasticsearch, Kibana, and ElastAlert — deployed through Security Onion, which also provides Zeek for network traffic analysis and Suricata for intrusion detection
Container Platform	Docker
Security Automation Engine	n8n
Case Management System	TheHive
Database Backend	Cassandra (for TheHive)
Firewall and Enforcement	FortiGate (REST API integration for automated IP and domain blocking)
Threat Intelligence Feeds	AbuseIPDB, VirusTotal, GreyNoise, URLScan.io
Attack Simulation	Hydra
Notification Channels	Telegram Bot API and Gmail

3.4.2 Hardware Specifications

Component	Specification
CPU	Intel Core i5-8250U 1.60 GHz
RAM	32 GB
Storage	150 GB SSD
Network	Stable internet connection required for threat intelligence API lookups

Code of Ethics (Project-Specific – Virtual SOC Project)

This project follows a strict Code of Ethics aligned with cybersecurity professional standards to ensure responsible, legal, and safe use of all tools, data, and simulations within the Virtual Security Operations Center (SOC) environment.

1. Authorized Use Only

All security tools, attack simulations, and detection mechanisms used in this project are strictly restricted to controlled, isolated, and authorized laboratory environments. No testing, scanning, or exploitation activities are performed against any real-world or unauthorized systems.

2. Data Privacy and Protection

No real user data, personal information, or production system logs are used within this project. All datasets, alerts, and network traffic are simulated or anonymized to ensure full compliance with privacy and data protection principles.

3. Responsible Security Testing

All simulated attack scenarios (such as brute-force attempts, phishing simulations, and malware indicators) are conducted solely for educational and validation purposes. These simulations are designed to improve detection and response capabilities and are never intended for harm or misuse.

4. Non-Misuse of Security Tools

Tools such as Suricata, Zeek, Elastic Stack, n8n, and FortiGate APIs are used exclusively for defensive security research and SOC workflow automation. They are not used for unauthorized access, disruption, or surveillance of external systems.

5. Ethical Handling of Threat Intelligence

External threat intelligence sources (e.g., AbuseIPDB, VirusTotal, GreyNoise, URLScan.io) are accessed in compliance with their terms of service. Retrieved data is used only for enrichment and analysis within the SOC pipeline.

6. Transparency and Academic Integrity

All results, findings, and implementations in this project are documented honestly and accurately without fabrication or misrepresentation. Any limitations or assumptions are clearly stated.

7. Compliance with Legal and Academic Standards

The project adheres to applicable cybersecurity laws, institutional policies, and academic integrity regulations. No activity in this project violates local or international cybersecurity legislation.

8. Defensive Purpose Only

The entire SOC architecture is designed strictly for defensive cybersecurity operations, including detection, monitoring, enrichment, and incident response. Offensive capabilities are simulated only within a controlled testing scope.

3.6 Analysis Diagrams

3.6.1 Use Case Diagram

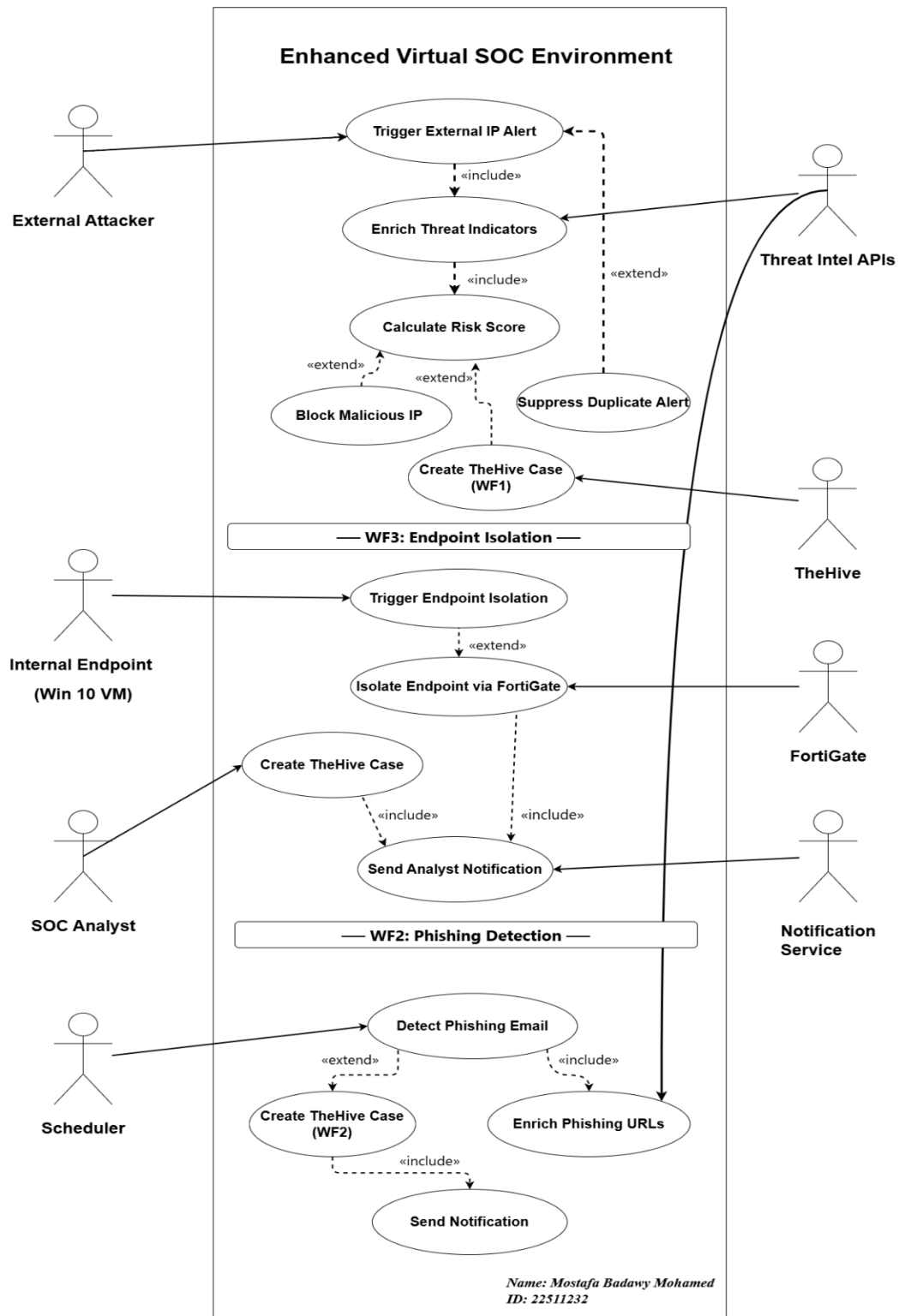


Figure 5 - use case diagram

3.6.2 Sequence diagram

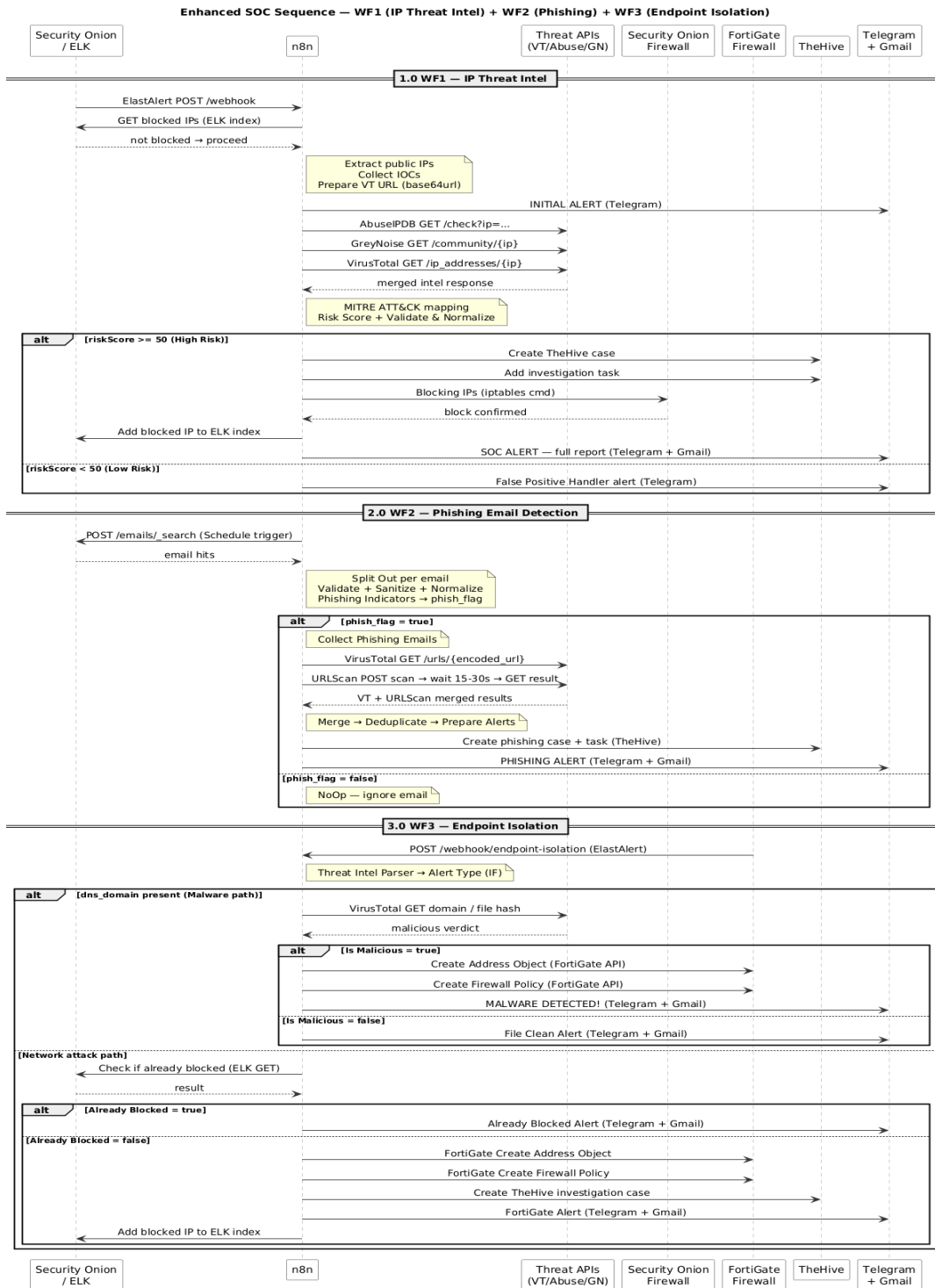


Figure 7 – sequence diagram

3.6.3 State Diagram

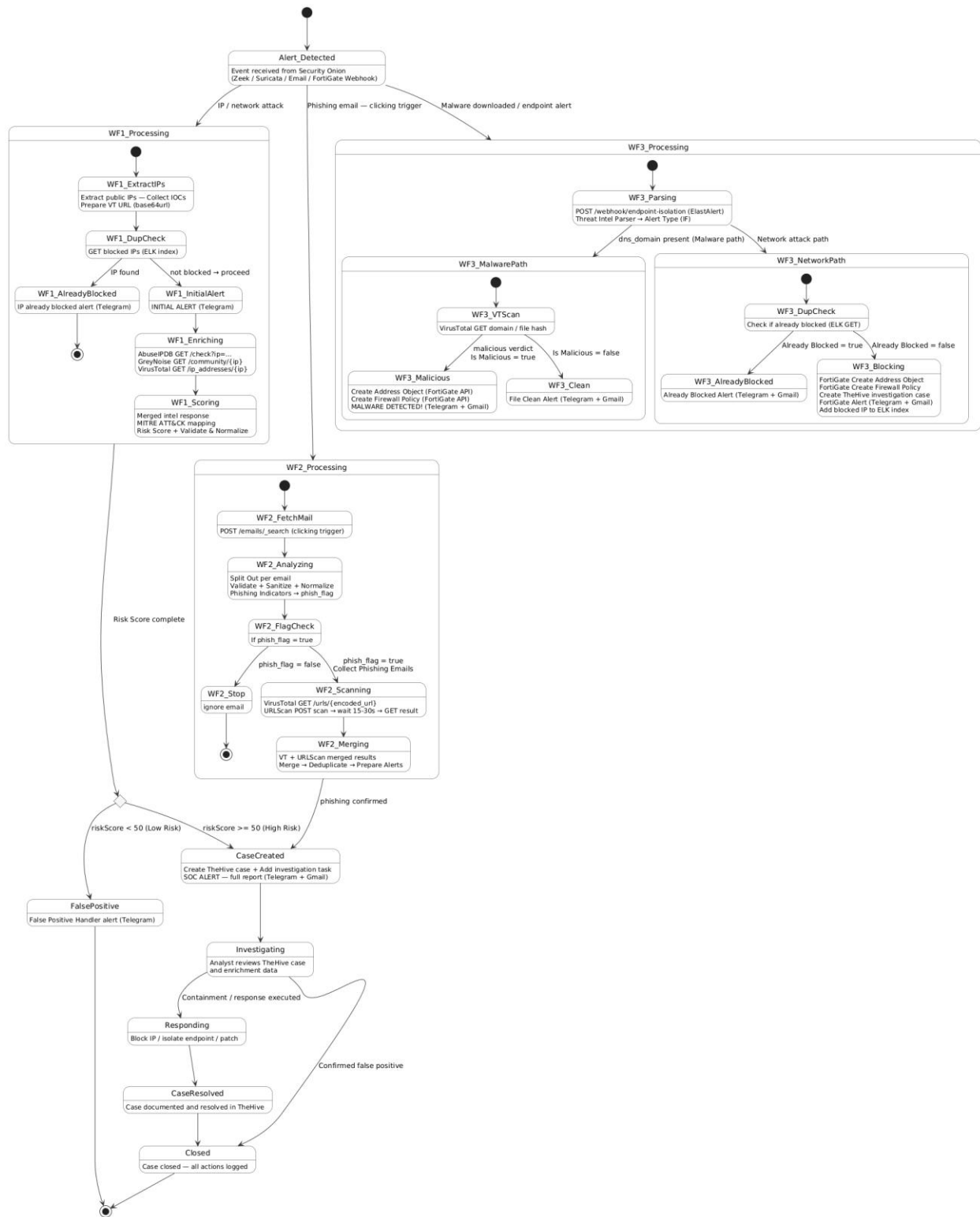


Figure 8— state diagram

3.6.4 Block Diagram

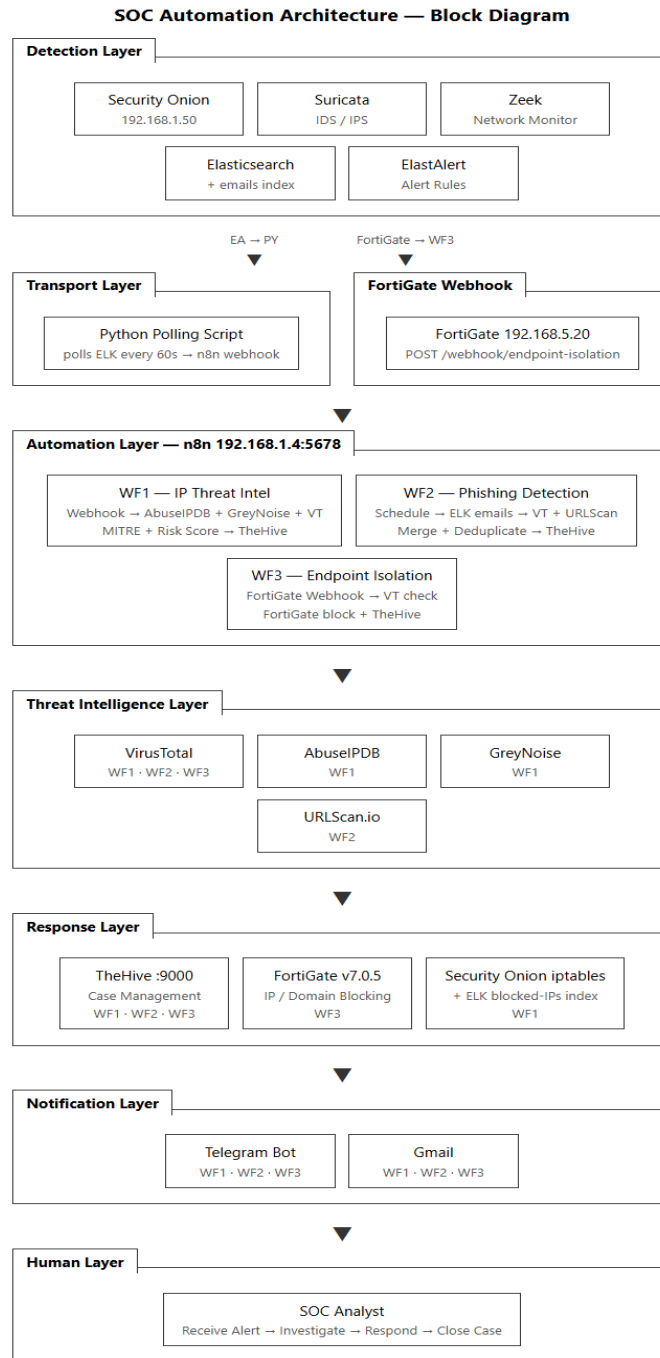


Figure 9 – block diagram

3.6.5 Activity Diagram

3.6.5.1 Activity Diagram WF1

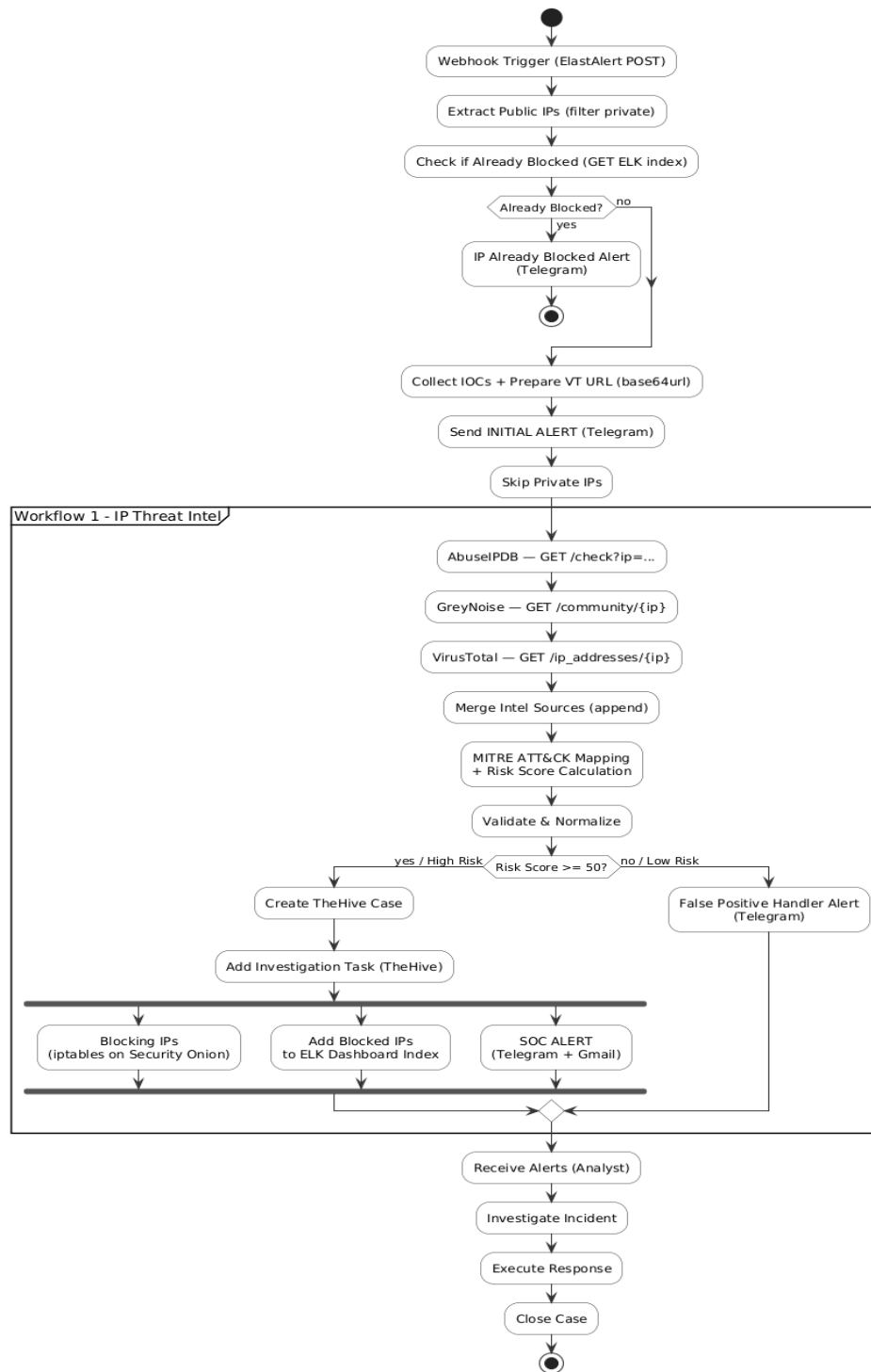


Figure 10 – WF1 activity Diagram

3.6.5.2 Workflow 2 - Phishing Detection

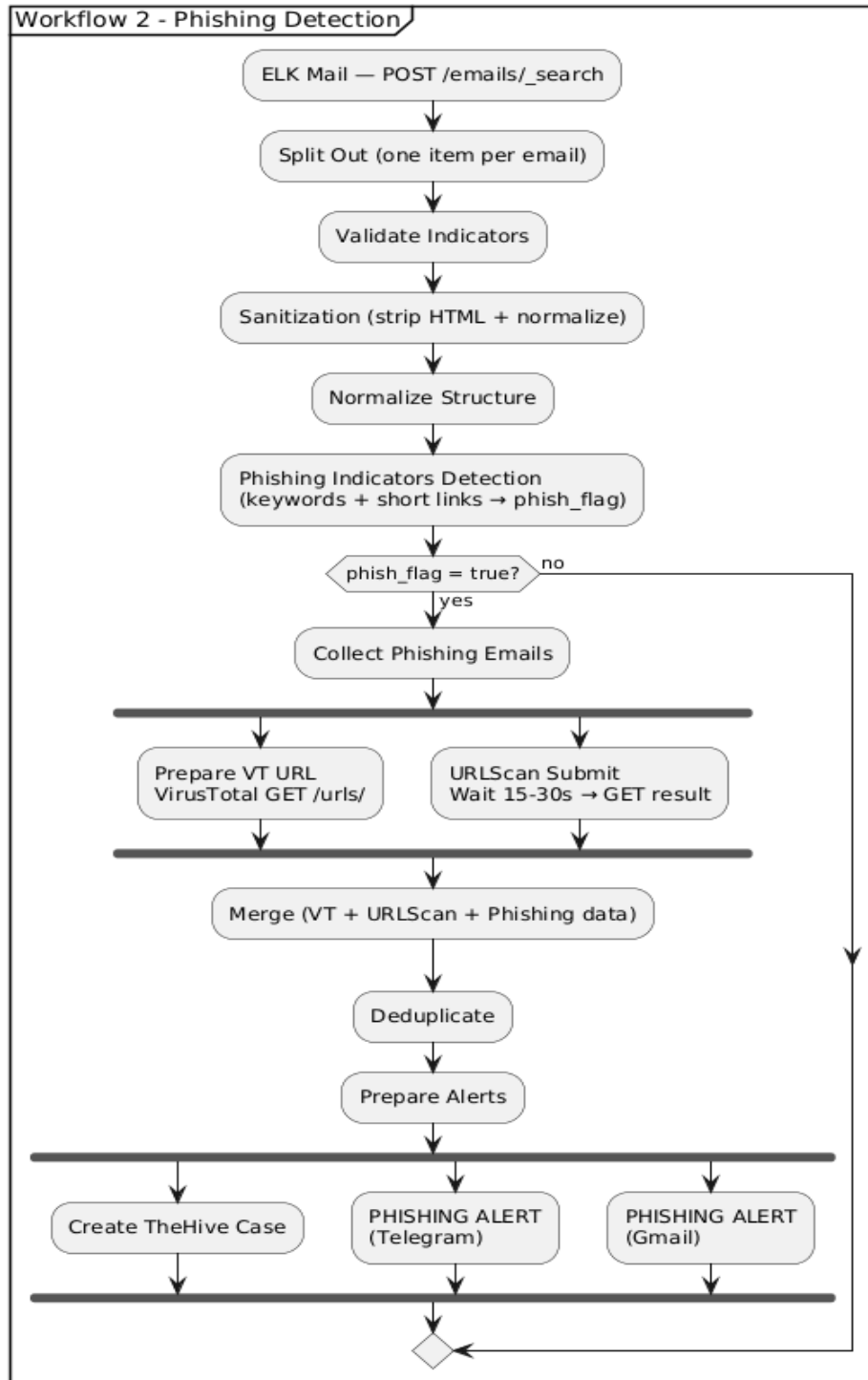


Figure 11 – WF2 activity Diagram

3.6.5.3 Activity WF3 Endpoint Isolation

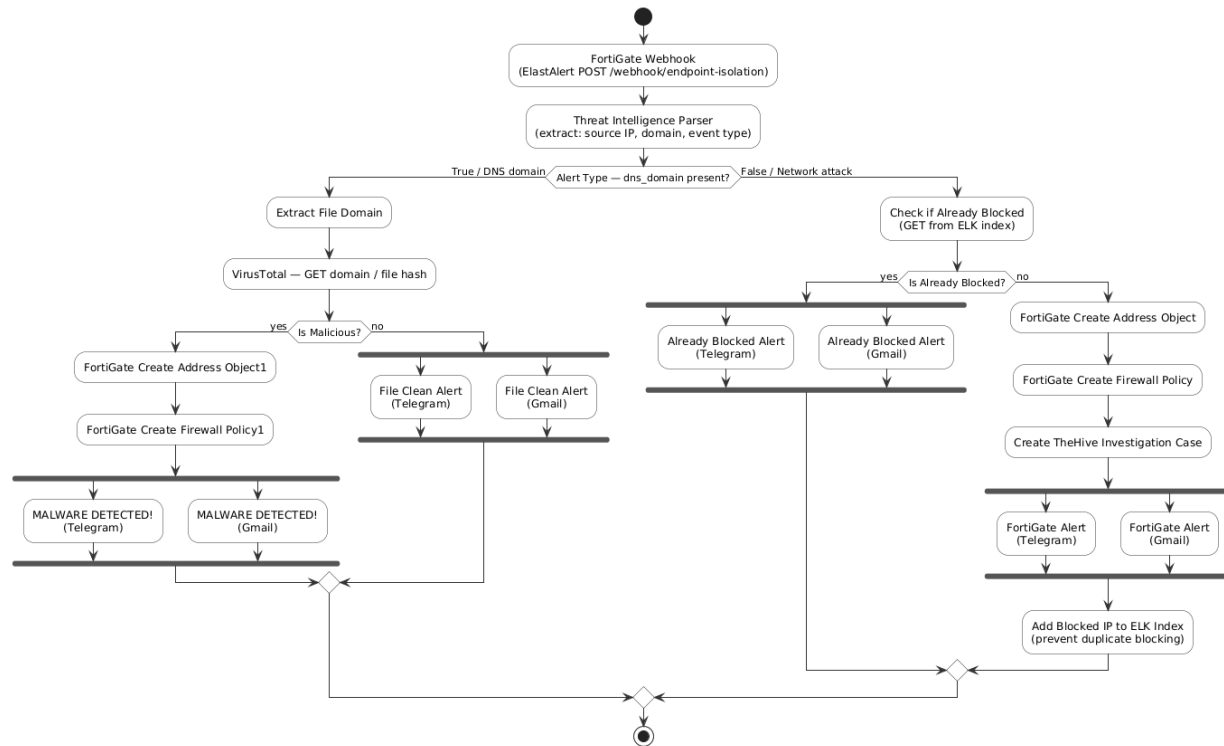


Figure 12 – WF3 activity Diagram

Chapter Four:

Design, Implementation and Testing

4.1 System Architecture

The virtual SOC is designed as a layered architecture in which detection, enrichment, automation, and enforcement operate as loosely coupled but tightly integrated services. Data flows from the network capture layer through the enrichment layer to the response layer, with notifications dispatched at each significant decision point.

The Elastic Stack forms the analytical core of the architecture. Network traffic captured from the monitored segment is processed by Zeek and Suricata running within Security Onion, and the resulting events are stored in Elasticsearch. Kibana provides dashboard visibility into detection activity and blocked indicator records. ElastAlert continuously evaluates stored events against predefined detection rules and, when a rule fires, constructs a JSON alert payload and forwards it to the appropriate n8n webhook endpoint.

The automation layer processes incoming alerts through three independent workflow tracks:

- Workflow 1 for Incident Management and IP Threat Intelligence.
- Workflow 2 for Phishing Email Detection.
- Workflow 3 for Endpoint Isolation and Malware Containment.

Each workflow communicates with:

- External threat intelligence APIs where applicable.
- TheHive for case creation.
- FortiGate for enforcement actions.
- Telegram and Gmail for analyst notifications.

All blocking decisions are recorded back into Elasticsearch to enable duplicate suppression in subsequent workflow cycles.

FortiGate sits at the enforcement boundary and is used exclusively by Workflow 3. When an internal endpoint threat is confirmed, n8n issues REST API calls to create an address object and apply a deny firewall policy, isolating the endpoint at the network level.

4.2 Infrastructure Setup

The system is deployed across two primary virtual machines and a set of Docker containers within a local lab network.

Security Onion Environment

Security Onion runs on a dedicated virtual machine and hosts the complete Elastic Stack alongside Zeek and Suricata. ElastAlert runs as a service within Security Onion and is configured with detection rules for:

- SSH brute-force activity.
- Potential port scans.
- DNS-based malware delivery patterns.

Kibana dashboards provide real-time visibility into stored events, risk scores, and blocked indicator records.

Automation and Response Layer

The automation and response layer runs n8n and TheHive as Docker containers.

- n8n receives webhook deliveries from ElastAlert.
- n8n issues outbound API requests to threat intelligence services, TheHive, and FortiGate.
- TheHive uses Cassandra as its backend database.
- TheHive is accessible to n8n via REST API for case creation and task assignment.

FortiGate Firewall

FortiGate operates as a virtual firewall on the lab network. Its REST API is configured to accept authenticated requests from n8n, allowing the automation layer to:

- Programmatically create address objects.
- Apply deny policies when endpoint isolation is required.

Endpoint Environment

A Windows 10 endpoint is connected to the monitored network segment. It serves as the target for:

- Atomic Red Team attack simulations.
- Hydra-based brute-force tests.

These activities generate realistic security telemetry for the detection pipeline.

4.3 Workflow Implementation

4.3.1 Workflow 1: Threat Detection and Enrichment

Workflow 1 is triggered via an HTTP webhook. During testing, the webhook was invoked manually using curl from a Kali Linux virtual machine, with the client IP field configured to represent known malicious external addresses including Tor exit nodes in order to simulate real ElastAlert-triggered security events. Multiple test iterations were performed using different source IP addresses to validate the duplicate suppression and repeat-attacker handling mechanisms implemented within the workflow.

The webhook payload is received by n8n, and a JavaScript code node extracts public IP addresses while filtering out RFC 1918 private ranges to ensure only routable, external indicators are processed.

Before enrichment begins, an Elasticsearch lookup checks whether the source IP has been acted in a previous cycle. If a matching record is found:

- The workflow dispatches a Telegram notification confirming the existing block.
- The workflow terminates without re-issuing duplicate API calls.

For new indicators:

- An initial Telegram alert is sent to inform the analyst that investigation is underway.
- Parallel HTTP requests are dispatched to AbuseIPDB, VirusTotal, and GreyNoise.
- A Merge node consolidates the three responses into a single item for scoring.

A MITRE ATT&CK-informed risk scoring node computes a composite score weighted across:

- VirusTotal malicious engine detections.
- AbuseIPDB confidence percentage.
- GreyNoise classification.
- Tor exit node status.

High-risk indicators are routed to the response branch:

- iptables blocking is applied on the Security Onion host.
- The blocked IP is recorded in Elasticsearch.
- A TheHive case is created with a full enrichment report and investigation task.
- SOC alerts are delivered via Telegram and Gmail.

Low-risk indicators are routed to a false positive handler that dispatches a low-priority notification without triggering any enforcement action.



Figure 13 – Workflow 1: Threat Detection and Enrichment

4.3.2 Workflow 2: Phishing Email Detection

Workflow 2 runs on a scheduled trigger and processes email records stored in an Elasticsearch index.

An HTTP POST request retrieves unprocessed records, and a Split-Out node separates each email into an individual processing item.

The workflow includes:

- Validation checks confirming the presence of required fields.
- A sanitization step removing HTML markup and non-ASCII characters.
- A normalization node reshaping each record into a consistent JSON structure.

A phishing indicator detection node evaluates each email against heuristic rules:

- Urgency keywords in the subject or body such as "verify", "suspend", or "reset".
- The presence of URL shortening services including bit.ly, tinyurl.com, and ow.ly.
- Embedded redirect links.

Emails satisfying one or more criteria receive a phishing flag and are routed to the enrichment branch; clean emails are discarded.

Flagged emails have their indicator URLs submitted in parallel to:

- VirusTotal using base64url-encoded URL lookup.
- URLScan.io.

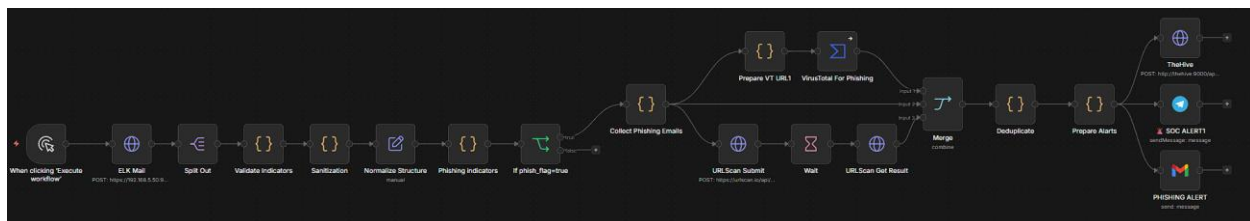
A wait node allows URLScan.io processing time before a GET request retrieves the full scan result including:

- Destination IP.
- Country.
- Verdict score.

A Merge node consolidates:

- The VirusTotal result.
- The URLScan.io result.
- The original email data.

TheHive receives a case creation request containing all enrichment fields and a severity classification. Telegram and Gmail alerts are dispatched simultaneously to the analyst.



This figure illustrates how email data is processed and analyzed to detect phishing attempts.

Figure 13 – 2: Phishing Email Detection

4.3.3 Workflow 3: Endpoint Isolation and Malware Containment

Workflow 3 is concerned with internal endpoint threats rather than external actor profiling.

Where Workflow 1 asks:

"Is this external IP malicious globally?"

Workflow 3 asks:

"Is this internal host compromised or actively attacking the network?"

The workflow responds by isolating the endpoint at the firewall level immediately.

The workflow is triggered by two ElastAlert rules:

1. Zeek DNS logs reveal that an internal endpoint is resolving domains associated with malware delivery.
2. Suricata detects that an internal endpoint is the source of an SSH brute-force attack directed at other hosts on the network segment.

The webhook payload is received by a FortiGate Webhook node.

A Threat Intelligence Parser:

- Extracts alert fields.
- Determines the indicator type.

An Alert Type branching node routes execution:

- If a DNS domain field is present, the flow enters the malware containment path.
- Otherwise, it enters the network attack isolation path.

Malware Containment Path

The queried domain is submitted to VirusTotal.

If the verdict is malicious:

- n8n calls the FortiGate REST API to create an address object.
- A deny firewall policy is applied for the domain.
- Telegram and Gmail MALWARE DETECTED alerts are dispatched confirming that endpoint isolation has been applied.

If the domain is clean:

- A File Clean alert is sent.
- No enforcement action is taken.

Network Attack Isolation Path

Elasticsearch is queried first to determine whether the source endpoint has already been blocked on FortiGate in a previous cycle.

If found:

- A Repeat Attacker alert is dispatched.
- The workflow terminates without issuing duplicate API calls.

If not found:

- n8n calls the FortiGate REST API to create an IP address object.
- A deny policy is applied, isolating the endpoint from the network until manual review.
- A FortiGate blocking confirmation alert is dispatched via Telegram and Gmail.

Unlike Workflow 1, Workflow 3:

- Does not perform multi-source threat intelligence enrichment.
- Does not create TheHive cases.

The ElastAlert rule match itself is sufficient justification for immediate isolation — the fact that the endpoint is internally generating attack traffic or resolving malware domains requires no further external verification before enforcement is applied.

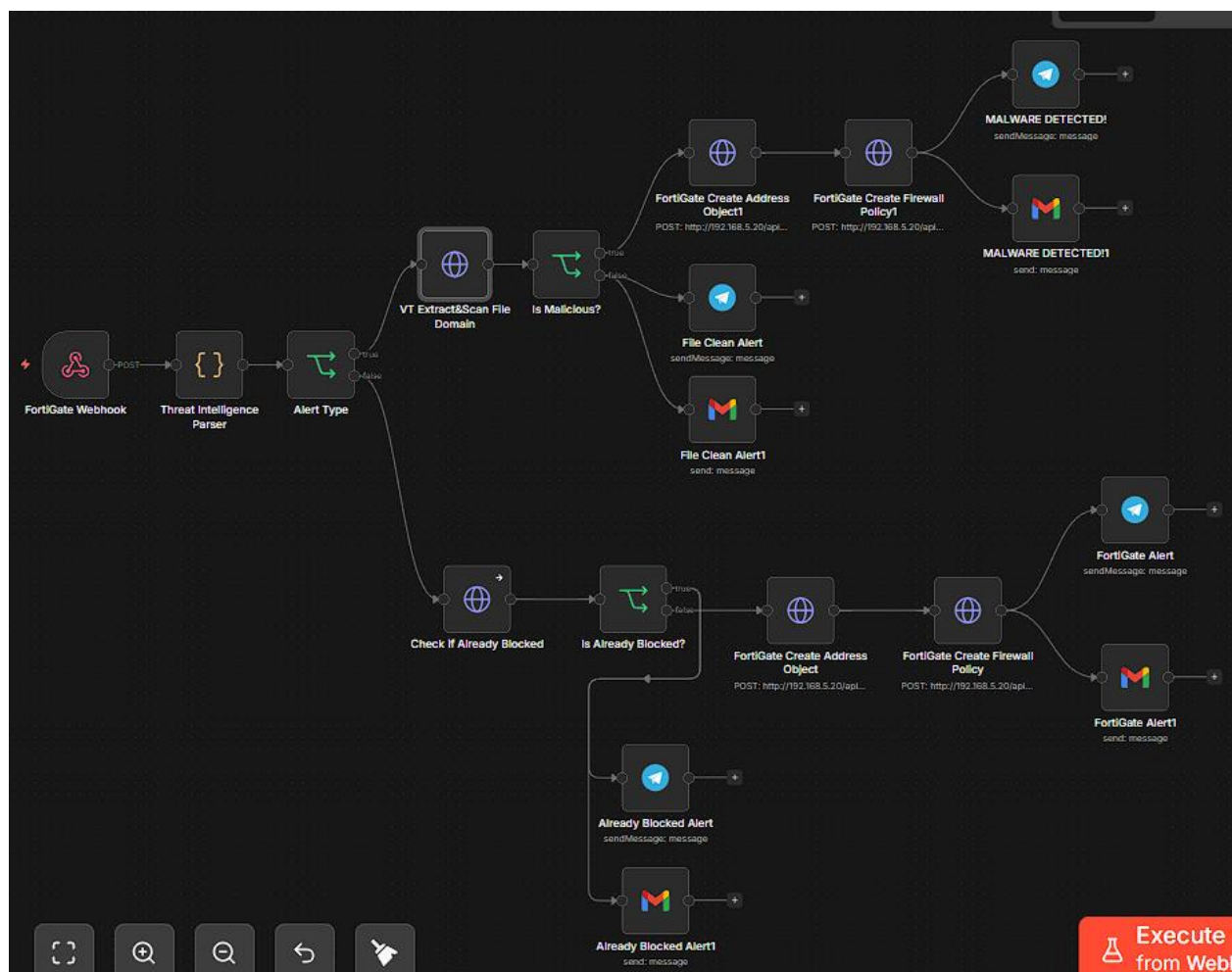


Figure 14 – Endpoint Isolation and Malware Containment

4.4 System Integration

All components communicate through authenticated API calls and HTTP webhooks.

Event and Alert Flow

- The Elastic Stack receives events from Zeek and Suricata.
- ElastAlert delivers alerts to n8n via webhooks.

External Integrations

n8n communicates with:

- TheHive.
- FortiGate.
- AbuseIPDB.
- VirusTotal.
- GreyNoise.
- URLScan.io.

All communication occurs through their respective REST APIs.

State Tracking

Blocking records are written back to Elasticsearch by n8n to maintain state across workflow cycles.

Notifications

Telegram and Gmail notifications are dispatched through their respective bot and SMTP APIs.

This architecture ensures that:

- Each component remains independently operable.
- Failure of any single external service produces a handled error notification rather than a silent workflow failure.

4.5 Challenges and Solutions

Challenge	Solution
Network communication between Docker containers and Security Onion	Firewall rule configuration was adjusted to permit webhook traffic while restricting unnecessary exposure
Inconsistent data formatting across threat intelligence API responses	A dedicated normalization node standardizes field names and data types before scoring
Duplicate enforcement actions	Eliminated through the Elasticsearch-based state tracking mechanism introduced in all three workflows
FortiGate REST API integration	Required precise authentication header configuration and correct JSON payload structure for address object creation and firewall policy application

Chapter Five:

Results and Discussion

5.1 Overview of Findings

The virtual SOC system demonstrated end-to-end functionality across all three workflows. Security events generated by controlled attack simulations were:

- Detected by the Elastic Stack via Zeek and Suricata.
- Forwarded to n8n via ElastAlert.
- Enriched through external threat intelligence APIs where applicable.
- Resolved through automated response actions.

All operations were completed within a timeframe consistent with near real-time operation.

TheHive cases were created automatically for confirmed threats in:

- Workflow 1.
- Workflow 2.

Telegram and Gmail notifications were delivered within seconds of each detection event across all three workflows.

5.2 Goals Achieved

All primary objectives defined in Chapter One were met.

Successfully Achieved Components

- The Elastic Stack was successfully deployed through Security Onion as the core SIEM platform.
- Three n8n workflows were implemented and validated against live simulated attack traffic.
- TheHive case management was confirmed to receive correctly structured case data from Workflow 1 and Workflow 2.
- FortiGate enforcement was validated by confirming that address objects and deny policies were created automatically by Workflow 3.
- Threat intelligence enrichment was confirmed for AbuseIPDB, VirusTotal, GreyNoise, and URLScan.io.

Notification Validation

Telegram and Gmail notifications were verified across all alert types:

- Initial alert.
- Full SOC alert.
- False positive.
- Already-blocked.
- Repeat attacker.
- Malware detected.
- File clean.

5.3 Testing Scenarios

Four test scenarios were executed to validate the system across all three workflows. Each scenario targets a specific threat category and confirms that detection, enrichment, response, and notification functions operate correctly end-to-end.

5.3.1 Workflow 1: IP Threat Intelligence and Incident Management

A brute-force attack was simulated using Hydra from a Kali Linux VM targeting the Windows 10 endpoint over SSH. The Elastic Stack detected the attack through Suricata signatures and generated a Critical-severity alert visible in the Kibana Security Alerts dashboard. ElastAlert forwarded the event to Workflow 1 for external IP analysis.

Workflow 1 extracted the external source IP and confirmed it had not been previously actioned. An initial Telegram alert was dispatched to notify the analyst that investigation was underway. The IP was then submitted in parallel to AbuseIPDB, VirusTotal, and GreyNoise. The enrichment returned a composite Risk Score of 100/100: the IP was flagged as malicious by 14 VirusTotal engines, carried a 97% AbuseIPDB abuse confidence score, and was identified as a Tor exit node — confirming it as a high-risk external threat actor.

The high-risk routing branch executed the following response actions:

- iptables blocking was applied on the Security Onion host
- the blocked IP was recorded in Elasticsearch for future duplicate suppression
- a TheHive case was created containing the complete Threat Intelligence Report and an investigation task with analyst next steps
- full SOC alerts were delivered via both Telegram and Gmail within seconds of the initial detection

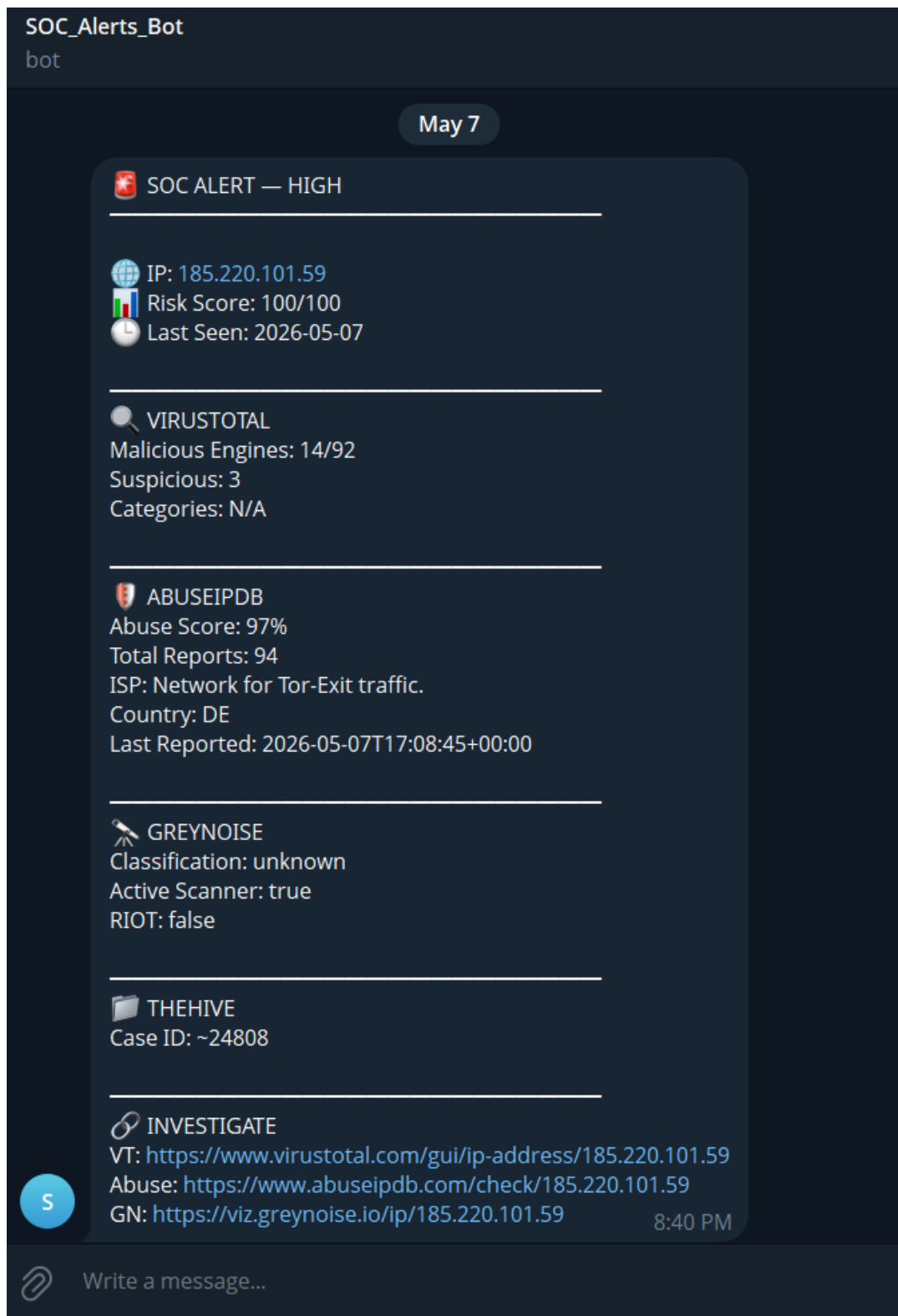


Figure 5 – Telegram SOC Alert (WF1)

42

5.3.2 Workflow 2: Phishing Email Detection

A simulated phishing email was injected into the Elasticsearch emails index to validate the phishing detection pipeline. The email was constructed to represent a realistic credential-harvesting attempt: it appeared to originate from `security@chase-bank.net`, carried the subject line "Urgent: Your Chase account suspended", contained a `bit.ly` shortened URL, and included urgency keywords matching the heuristic detection criteria configured in Workflow 2.

Workflow 2 retrieved the email on its scheduled trigger cycle. The sanitization and normalization stages processed the record successfully, and the phishing indicator detection node flagged the email based on the presence of a URL shortener and urgency keywords. The flagged URL was submitted in parallel to VirusTotal and URLScan.io. VirusTotal reported 15 malicious detections out of 93 scanning engines. URLScan.io returned a full scan result including the destination IP, country, and report URL.

A Merge node consolidated all enrichment data, and a TheHive case was created containing the complete phishing report — sender details, extracted IOCs, detection reasons, VirusTotal results, and the URLScan.io report link — with severity classified as HIGH. Telegram and Gmail Phishing Alerts were dispatched simultaneously to the analyst.

The screenshot displays the TheHive Case #1 interface. On the left, a sidebar shows a list of cases with columns for STATUS, #NUMBER, TITLE, and SEVERITY. The selected case is titled "Phishing Alert: Urgent: Your Chase account suspended" with a status of "New" and a severity of "None". The main panel on the right provides details for Case #1, including its ID (~40968368), creation date (07/05/2026 17:25), and assignee (analyst). It also shows the start and end dates (2026-05-07). The case is tagged with "phishing", "soc", "virustotal", and "nbn". The description field contains a detailed phishing report, including the sender (security@chase-bank.net), subject (Urgent: Your Chase account suspended), IOCs (http://bit.ly/chase-verify), reasons (verify, account suspended, short_link:bit.ly), VirusTotal results (15 malicious, 47 suspicious, 93 total), URLScan.io results (IP: 67.199.248.11, Country: US), and a report link (https://urlscan.io/result/019e02d2-751d-75df-b256-8e3bb5a0979c/). The severity is classified as HIGH. At the bottom right, there is a "Go to details" button.

Figure 6 – The Hive Case (WF2)

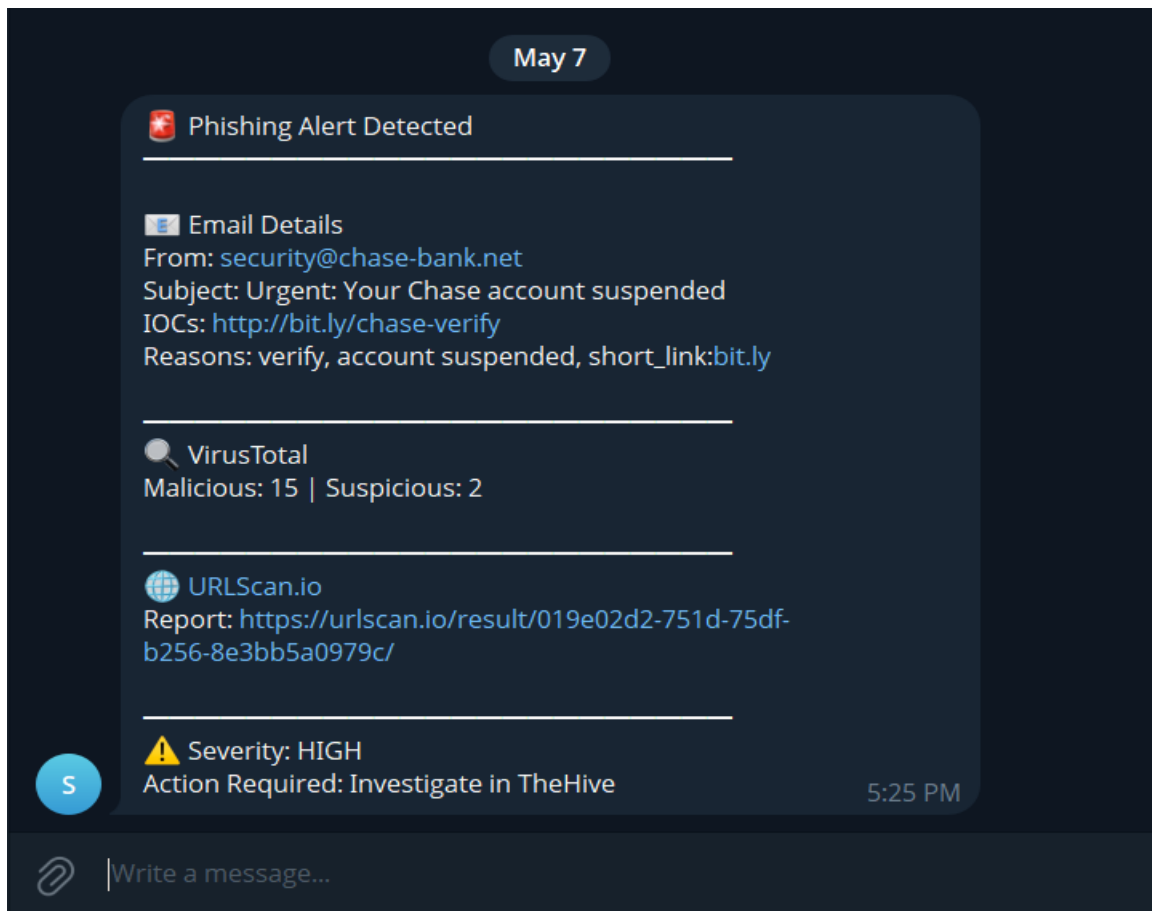
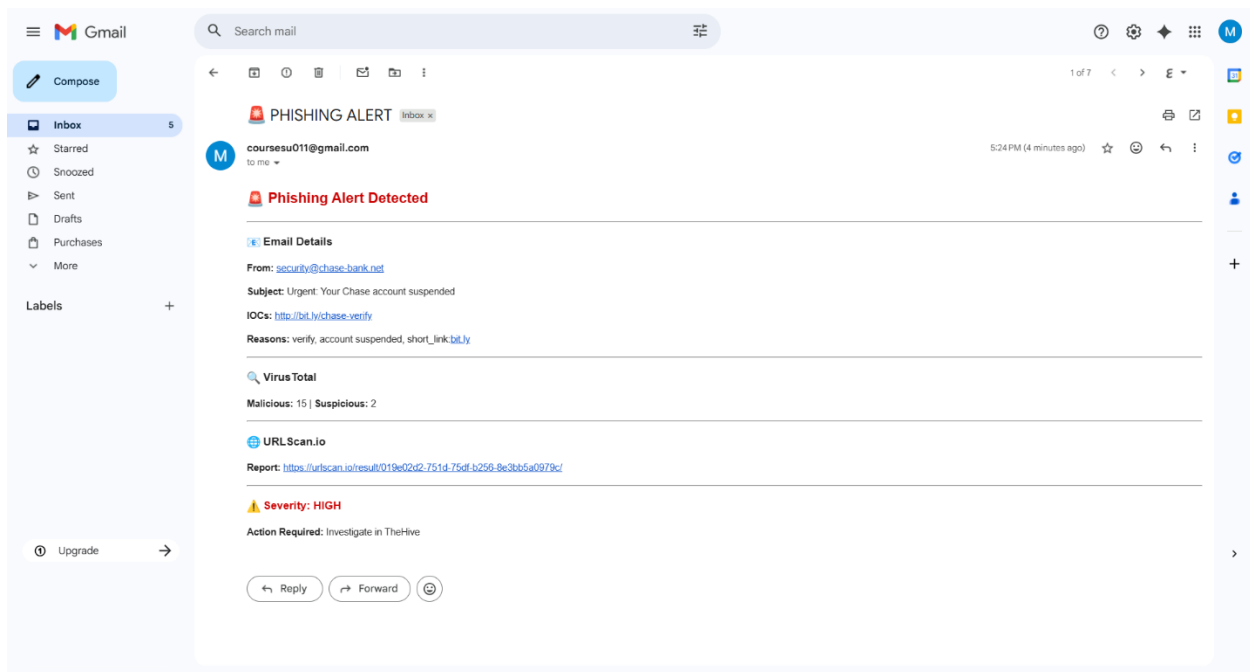


Figure 7 – Email and Telegram Alerts (WF2)

5.3.3 Workflow 3: Endpoint Isolation and Malware Containment

Workflow 3 was validated across its two execution paths independently.

5.3.3.1 Malware Containment Path

A malware simulation was performed by attempting to download the EICAR test file from eicar.org on the Windows 10 endpoint. Windows Defender flagged the download attempt, and Zeek DNS logs captured the DNS query to eicar.org. The Elastic Stack generated two High-severity Malware Download alerts visible in the Kibana Security Alerts dashboard. ElastAlert triggered Workflow 3 with a DNS domain field in the payload, routing execution into the malware containment path.

Workflow 3 submitted eicar.org to VirusTotal and confirmed the domain as malicious. n8n called the FortiGate REST API to create an address object for the domain and apply a deny firewall policy. The new policy entry was confirmed by inspecting the FortiGate Addresses and Firewall Policy tables. MALWARE DETECTED alerts were delivered via Telegram and Gmail confirming that endpoint isolation had been applied at the FortiGate level.

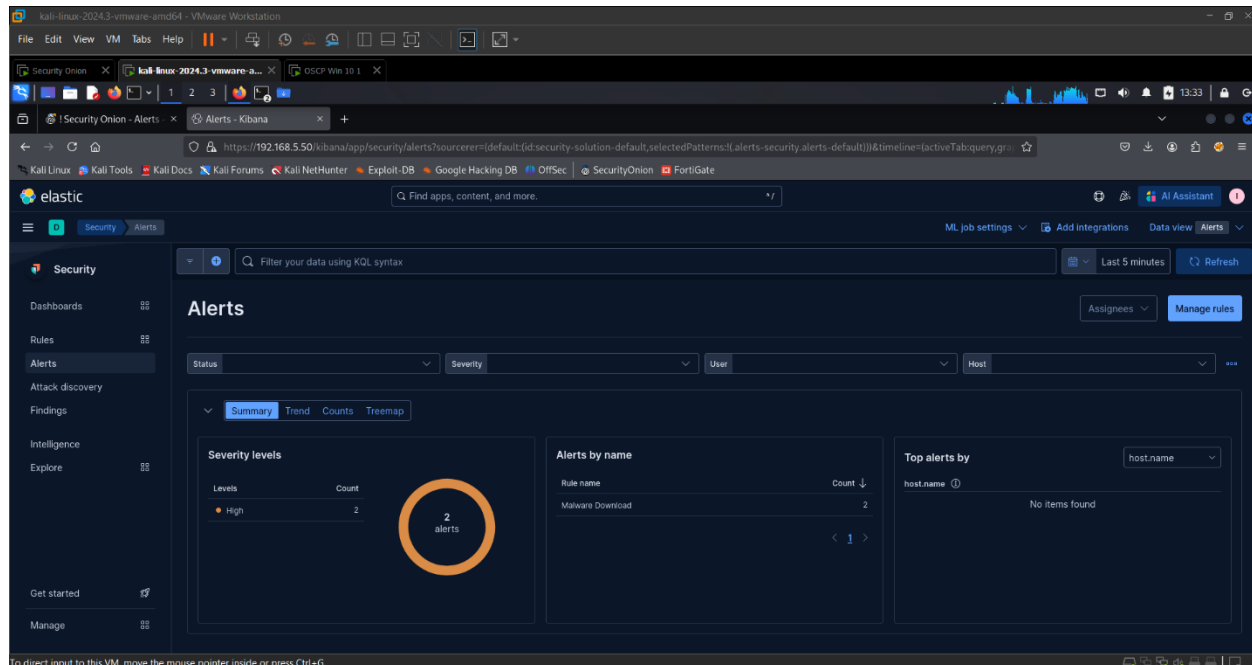


Figure 8 – Malware Alert in Kibana (WF2)

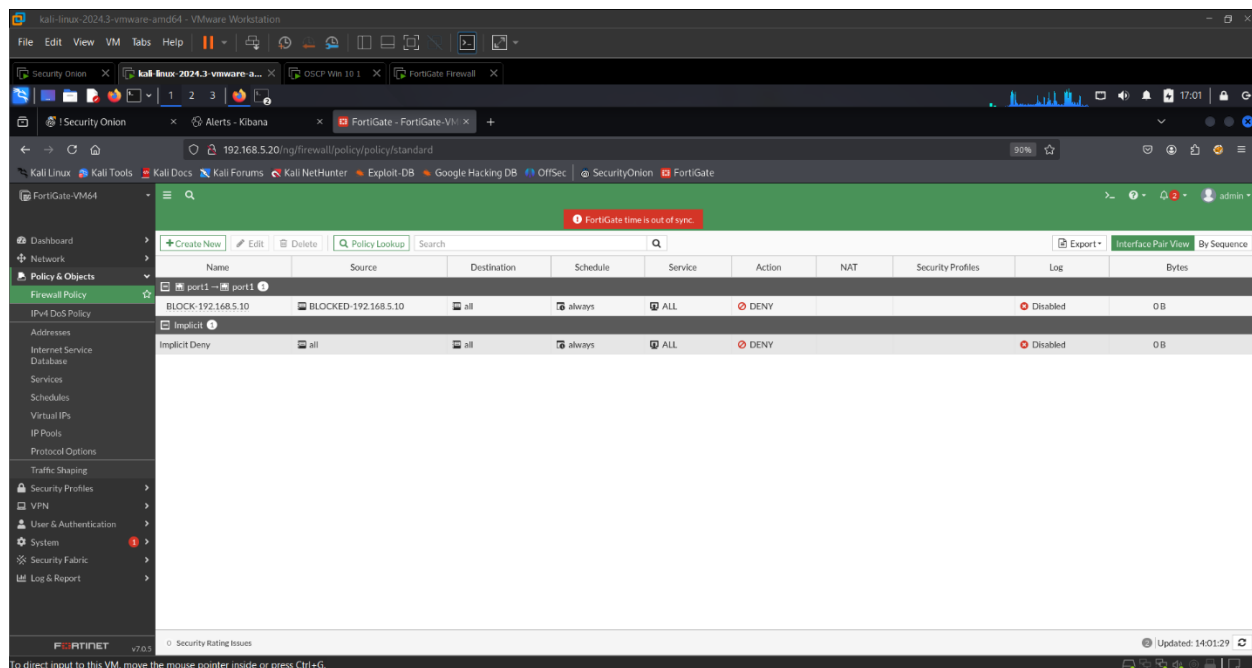


Figure 9 – FortiGate Policy (WF3)

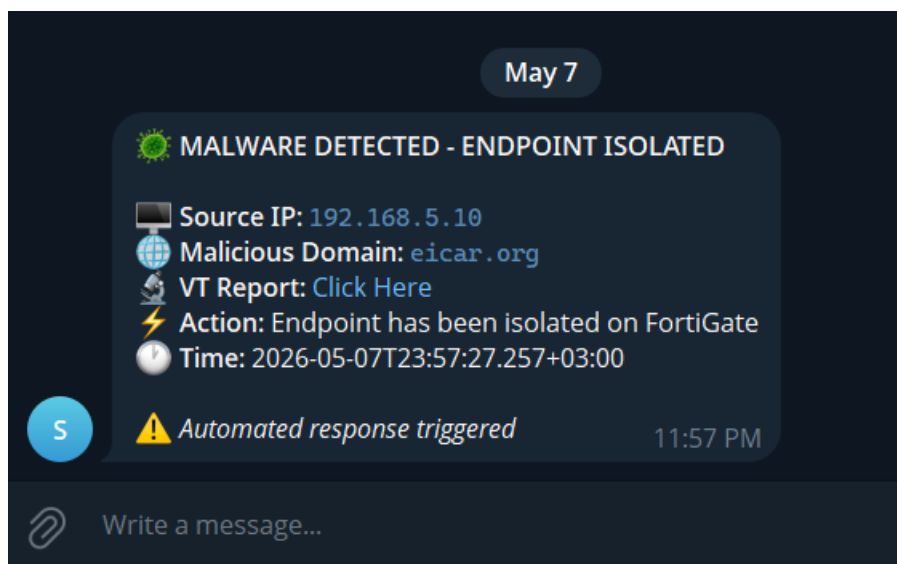


Figure 10 – Telegram MALWARE DETECTED Alert (WF3)

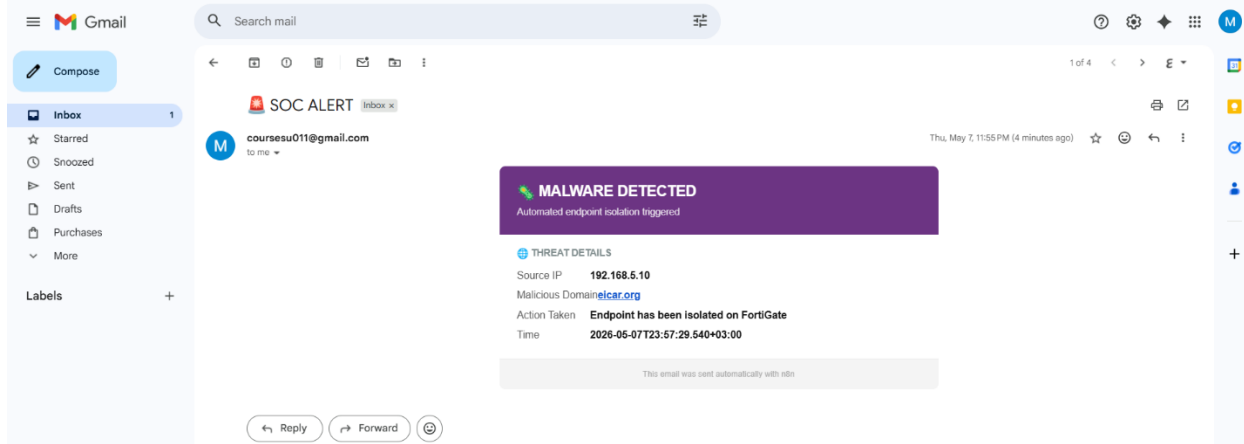


Figure 11 – Email MALWARE DETECTED Alert (WF3)

5.3.3.2 Network Attack Isolation Path

The same SSH brute-force simulation used in the Workflow 1 test was also processed by Workflow 3, which handled the internal dimension of the attack independently. ElastAlert forwarded the event with a source IP field rather than a DNS domain field, routing execution into the network attack isolation path. Workflow 3 queried Elasticsearch, confirmed the source endpoint had not been previously blocked on FortiGate, and called the FortiGate REST API to create an IP address object and apply a deny policy for the internal source endpoint. The block entry was confirmed in the FortiGate Firewall Policy table. A FortiGate blocking confirmation

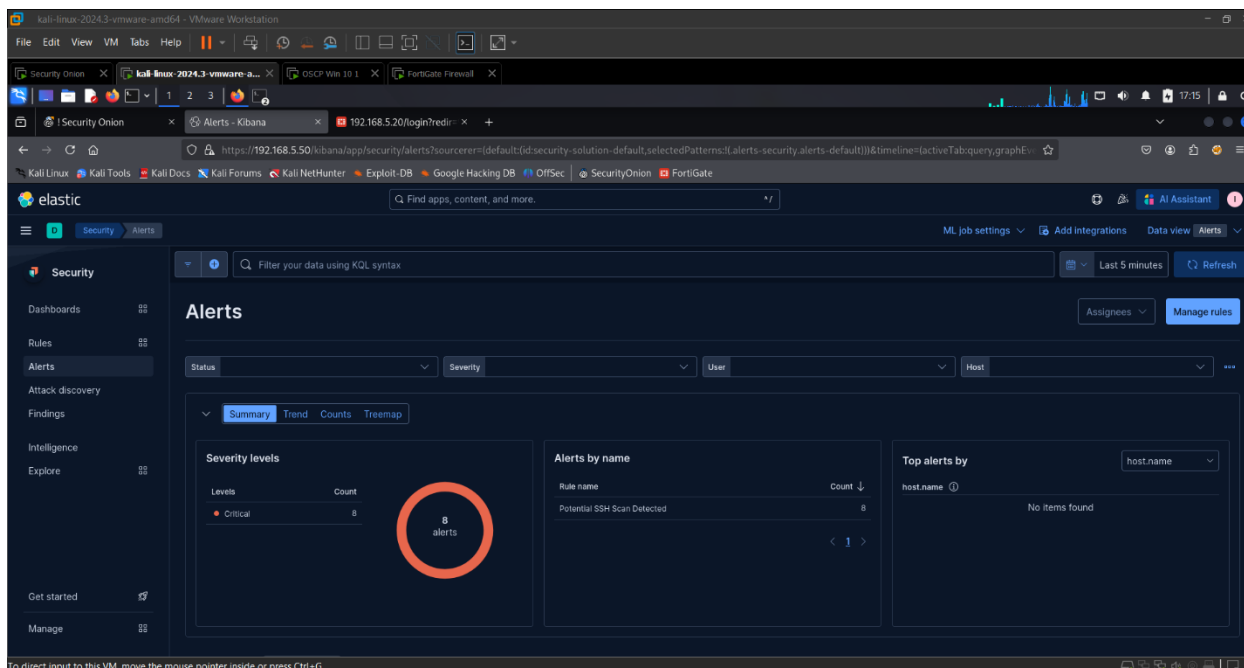


Figure 12 – SSH Scan Detected Alert (WF3)

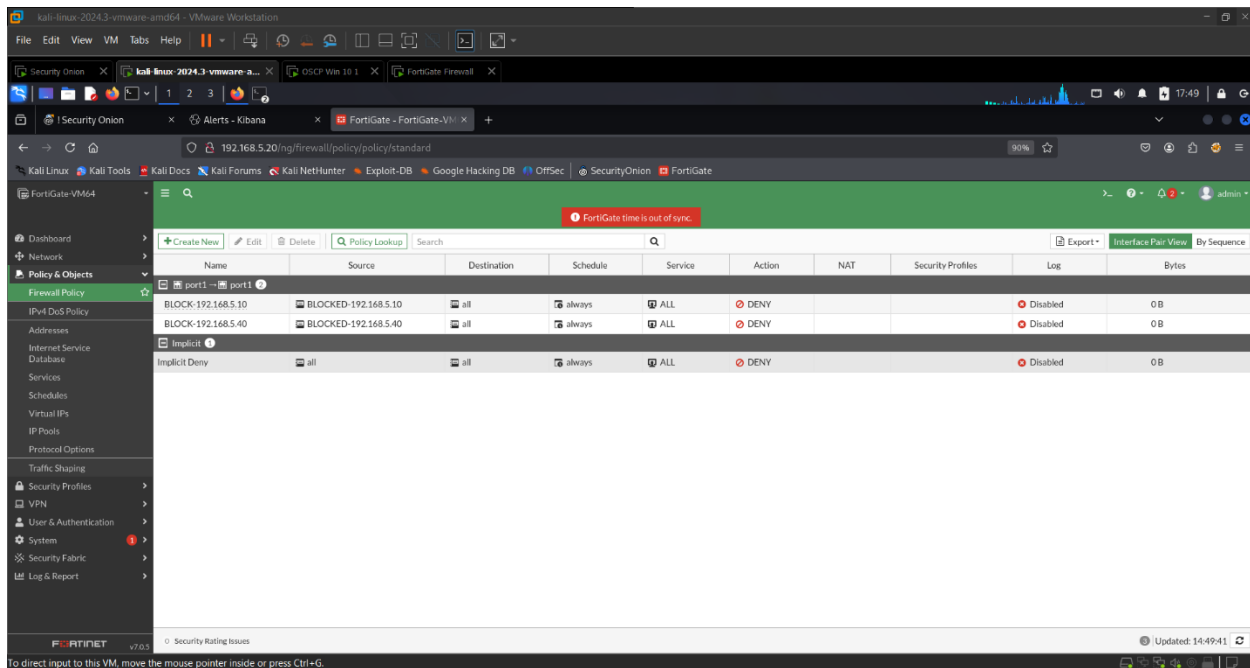


Figure 13 – FortiGate Policies (WF3)

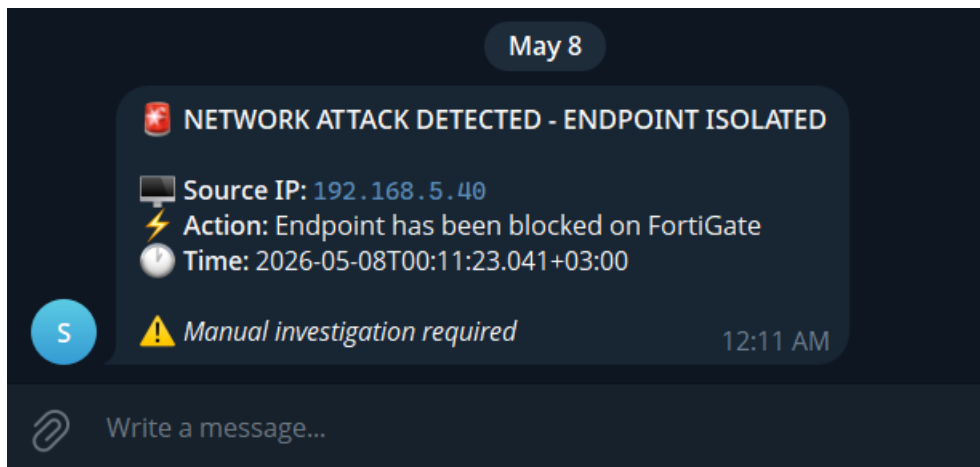


Figure 14 – Telegram NETWORK ATTACK DETECTED Alert (WF3)

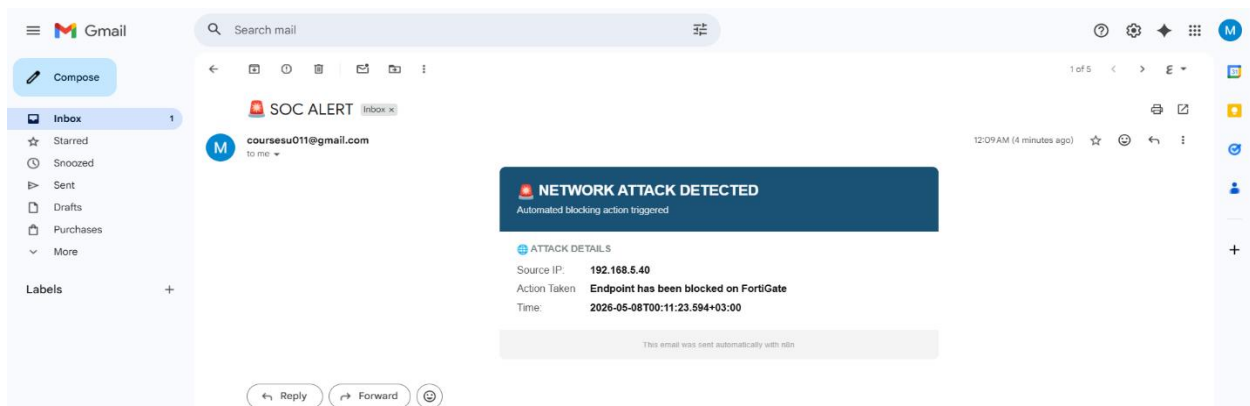


Figure 15 – Email NETWORK ATTACK DETECTED Alert (WF3)

5.3.4 Repeat Attacker Suppression

A second attack from a previously blocked source was simulated to validate the duplicate suppression mechanism.

Workflow 1 Validation

Workflow 1:

- Correctly identified the source IP as already present in the Elasticsearch blocked-IPs index.
- Terminated after dispatching a Repeat Attacker notification.

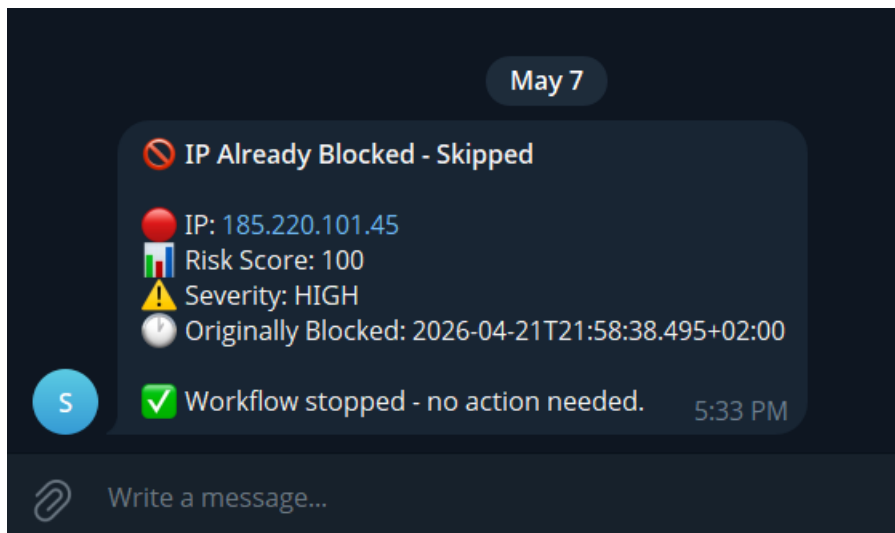


Figure 16 – Telegram IP Already Blocked Alert (WF1)

Workflow 3 Validation

Workflow 3:

- Correctly identified the source endpoint as already blocked on FortiGate.
- Terminated without issuing duplicate API calls.

Both confirmations validate that the state-tracking mechanism functions reliably across consecutive attack cycles.

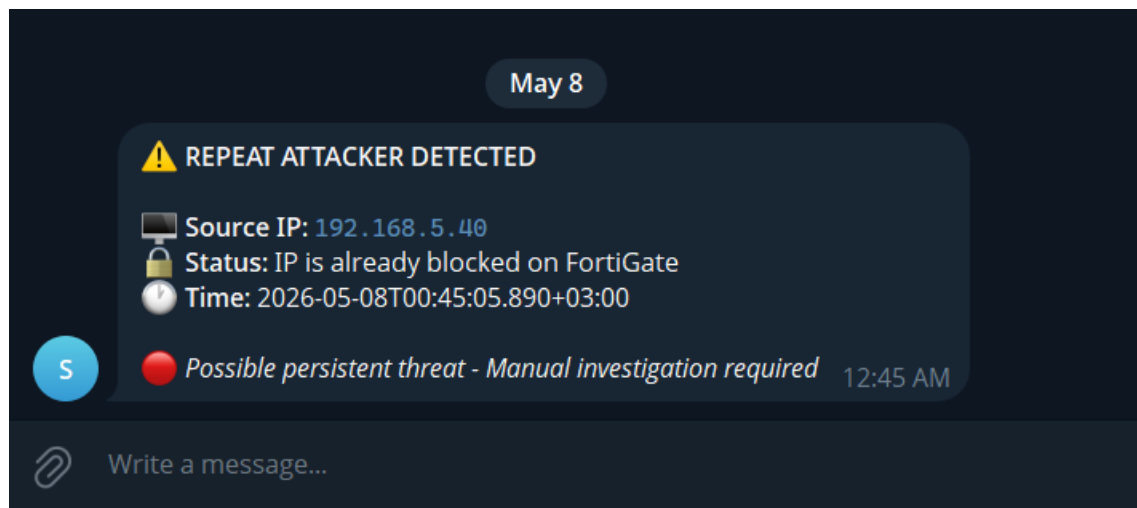


Figure 17 – Telegram REPEAT ATTACKER DETECTED Alert (WF3)

5.4 Performance Evaluation

Detection Accuracy

Detection accuracy was high across all tested scenarios:

- All simulated attack types triggered the correct detection rules.
- All confirmed threats were correctly classified.
- All threats were routed to the appropriate response branch.

Response Time

Response time — measured from ElastAlert trigger to Telegram alert delivery — was consistently within seconds for all three workflows, meeting the near real-time processing requirement.

Duplicate Suppression

Duplicate suppression was confirmed to function correctly across multiple consecutive test cycles.

FortiGate Validation

FortiGate policy creation was verified by inspecting:

- The Addresses table.
- The Firewall Policy table.

Verification was performed after each Workflow 3 execution.

5.5 Limitations

The system operates within a single-analyst, local virtualized environment and does not reflect the scale or data volumes of an enterprise SOC deployment.

Current Constraints

- Performance is constrained by available hardware.
- Running multiple workflows simultaneously against high-volume attack traffic would require additional compute resources.
- Enrichment lookups depend on external API availability.
- If a threat intelligence service is unreachable, the affected step times out and the workflow proceeds without that data source.
- Detection coverage is limited to threat types addressed by the configured Suricata and Zeek rules.

Advanced persistent threats and zero-day techniques are outside the current detection scope.

5.6 Future Work

Several enhancements are identified for future development.

Proposed Enhancements

- Machine learning-based anomaly detection could be layered on top of the existing rule-based system to extend coverage to unknown threat types.
 - Workflow 2 could be extended to process real inbound email through IMAP integration rather than relying on a manually populated Elasticsearch index.
 - Workflow 3 could be expanded to include dynamic VLAN assignment for isolated endpoints.
 - Deployment in a cloud environment would remove the hardware constraints of the current setup.
 - Integration with MISP for automated IOC sharing would further enhance the threat intelligence pipeline across all three workflows.
-

5.7 Ethical and Legal Considerations

All attack simulations were performed within an isolated virtual network using tools and techniques applied exclusively against systems owned and controlled for this project.

Ethical Compliance

- No external systems were targeted.
- No real user data was processed.
- The EICAR test file is a standard, harmless test artifact that does not execute malicious code.
- All open-source tools are used in accordance with their respective licenses.
- Threat intelligence API lookups were performed using publicly available services under their standard terms of use.

Chapter Six: Conclusion

6.1 Summary

This project delivered a fully functional virtual Security Operations Center built on open-source technologies and deployable on modest hardware.

The Elastic Stack, deployed through Security Onion, provides the detection and log management foundation.

Three dedicated n8n workflows:

- Workflow 1 for Incident Management and IP Threat Intelligence.
- Workflow 2 for Phishing Email Detection.
- Workflow 3 for Endpoint Isolation and Malware Containment.

These workflows automate the full response lifecycle from alert triage through enforcement and notification.

TheHive provides structured case management for confirmed threats, and FortiGate delivers network-level enforcement for internal endpoint threats.

6.2 Key Achievements

The project demonstrated that a production-representative SOC pipeline can be assembled without commercial licensing costs.

Major Achievements

- All three workflows performed reliably against live simulated attack traffic.
- Threats were correctly classified.
- Duplicate alerts were successfully suppressed.
- Structured TheHive cases were created automatically.
- Enforcement actions were executed on FortiGate and via iptables.
- Telegram and Gmail notifications were delivered within seconds of each detection event across all tested scenarios.

Beyond the technical outcomes, the project produced a documented and reproducible architecture that can serve as a foundation for further research and academic deployment.

6.3 Broader Significance

The gap between theoretical cybersecurity education and operational practice remains a significant challenge.

Tools that are standard in enterprise security programs are unavailable to most students and small organizations.

This project demonstrates that meaningful SOC capabilities — including:

- Automated enrichment.
- Case management.
- Network-level enforcement.

These capabilities are achievable within those constraints using mature, well-documented, and actively maintained open-source tooling.

6.4 Limitations and Future Directions

The current implementation is scoped to a single-analyst virtual environment, and several limitations follow from that scope:

- Hardware-constrained performance.
- Dependency on external API availability.
- Rule-based detection coverage.

Future Development Priorities

- Anomaly detection integration to extend coverage beyond known signatures.
 - Cloud deployment to remove hardware constraints.
 - Real-time email integration to replace the current manual index approach.
 - MISP integration for automated IOC sharing across all three workflows.
-

6.5 Final Remarks

This project confirms that open-source tooling has reached a level of maturity where a small team or individual practitioner can build and operate a SOC environment that replicates the core functions of enterprise security operations.

The system was:

- Built.
- Tested against live attack simulations.
- Validated end-to-end across three distinct threat categories.

That practical grounding — detection, enrichment, enforcement, and notification working together in a single automated pipeline — distinguishes this work from theoretical models and makes it a meaningful contribution to the field of cybersecurity education and open-source security operations.

References

Behl, A. and Behl, K. (2017) *Cybersecurity and Cyberwar: What Everyone Needs to Know*. Oxford: Oxford University Press.

Scarfone, K. and Mell, P. (2007) *Guide to Intrusion Detection and Prevention Systems (IDPS)*. Gaithersburg: National Institute of Standards and Technology (NIST). Available at: <https://nvlpubs.nist.gov>

Elastic (2024) *Elastic Stack Documentation*. Available at: <https://www.elastic.co/guide/index.html>

TheHive Project (2024) *TheHive Documentation*. Available at: <https://thehive-project.org>

n8n (2024) *n8n Workflow Automation Documentation*. Available at: <https://docs.n8n.io>

MITRE (2024) *MITRE ATT&CK Framework*. Available at: <https://attack.mitre.org>

Red Canary (2024) *Atomic Red Team Documentation*. Available at: <https://atomicredteam.io>

Palo Alto Networks (2024) *Cortex XSOAR Overview*. Available at: <https://www.paloaltonetworks.com/cortex/cortex-xsoar>

Splunk (2024) *Splunk Enterprise Security*. Available at: https://www.splunk.com/en_us/solutions/siem.html

Shodan (2024) *Shodan Search Engine*. Available at: <https://www.shodan.io>

AbuseIPDB (2024) *IP Reputation Database*. Available at: <https://www.abuseipdb.com>

VirusTotal (2024) *VirusTotal Intelligence*. Available at: <https://www.virustotal.com>

ENISA (2023) *Security Operations Centres (SOCs) Overview*. Available at: <https://www.enisa.europa.eu>

Behl, A. (2020) *Emerging Cybersecurity Technologies and Applications*. CRC Press.